Inductance and Resistance of Transmission Lines

2

2.1 INTRODUCTION

The four parameters which affect the performance of a transmission line as an element of a power system are inductance, capacitance, resistance and conductance. Shunt conductance, which is normally due to leakage over line insulators, is almost always neglected in overhead transmission lines. This chapter deals with the series line parameters, i.e., inductance and resistance. These parameters are uniformly distributed along the line and they together form the series impedance of the line.

Inductance is by far the most dominant line parameter from a power system engineer's viewpoint. As we shall see in later chapters, it is the inductive reactance which limits the transmission capacity of a line.

2.2 DEFINITION OF INDUCTANCE

Voltage induced in a circuit is given by

$$e = \frac{\mathrm{d}\psi}{\mathrm{d}t} \text{ V} \tag{2.1}$$

where ψ represents the flux linkages of the circuit in weber-turns (Wb-T). This can be written in the form

$$e = \frac{\mathrm{d}\psi}{\mathrm{d}i} \cdot \frac{\mathrm{d}i}{\mathrm{d}t} = L \frac{\mathrm{d}i}{\mathrm{d}t} \text{ V} \tag{2.2}$$

where $L = \dfrac{\mathrm{d}\psi}{\mathrm{d}i}$ is defined as the inductance of the circuit in henry, which in general may be a function of i. In a linear magnetic circuit, i.e., a circuit with constant permeability, flux linkages vary linearly with current such that the inductance is constant given by

$$L = \frac{\psi}{i} \text{ H}$$

or,

$$\psi = Li \text{ Wb-T} \tag{2.3}$$

If the current is alternating, the above equation can be written as

$$\lambda = LI \tag{2.4}$$

where λ and I are the rms values of flux linkages and current respectively. These are of course in phase.

Replacing $\dfrac{\mathrm{d}}{\mathrm{d}t}$ in Eq. (2.1) by $j\omega$, we get the steady state AC voltage drop due to alternating flux linkages as

$$V = j\omega LI = j\omega\lambda \text{ V} \tag{2.5}$$

On similar lines, the mutual inductance between two circuits is defined as the flux linkages of one circuit due to current in another, i.e.,

$$M_{12} = \frac{\lambda_{12}}{I_2} \text{ H} \tag{2.6}$$

The voltage drop in circuit 1 due to current in circuit 2 is

$$V_1 = j\omega M_{12} I_2 = j\omega \lambda_{12} \text{ V} \tag{2.7}$$

The concept of mutual inductance is required while considering the coupling between parallel lines and the influence of power lines on telephone lines.

2.3 FLUX LINKAGES OF AN ISOLATED CURRENT-CARRYING CONDUCTOR

Transmission lines are composed of parallel conductors which, for all practical purposes, can be considered as infinitely long. Let us first develop expressions for flux linkages of a long isolated current-carrying cylindrical conductor with return path lying at infinity. This system forms a single-turn circuit, flux linking which is in the form of circular lines concentric to the conductor. The total flux can be divided into two parts, that which is internal to the conductor and the flux external to the conductor. Such a division is helpful as the internal flux progressively links a smaller amount of current as we proceed inwards, towards the centre of the conductor, while the external flux always links the total current inside the conductor.

2.3.1 Flux Linkages due to Internal Flux

Figure 2.1 shows the cross-sectional view of a long cylindrical conductor carrying current I.

The mmf round a concentric closed circular path of radius y internal to the conductor as shown in the figure is

$$\oint H_y \cdot ds I_y \text{ (Ampere's law)} \tag{2.8}$$

where,

H_y = magnetic field intensity (AT/m)

I_y = current enclosed (A)

By symmetry, H_y is constant and is in the direction of ds all along the circular path. Therefore, from Eq. (2.8), we have

$$2\pi y H_y = I_y \tag{2.9}$$

Assuming uniform current density*

$$I_y = \left(\frac{\pi y^2}{\pi r^2}\right) I = \left(\frac{y^2}{r^2}\right) I \tag{2.10}$$

From Eqs (2.9) and (2.10), we obtain

$$H_y = \frac{yI}{2\pi r^2} \text{ AT/m} \tag{2.11}$$

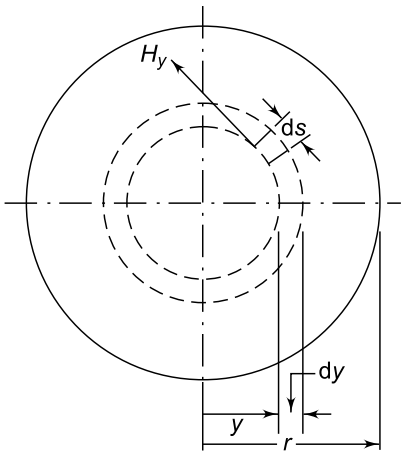


Fig. 2.1 *Flux linkages due to internal flux (cross-sectional view)*

* For power frequency of 50 Hz, it is quite reasonable to assume uniform current density. The effect of non-uniform current density is considered later in this chapter while treating resistance.

The flux density B_y, y metres from the centre of the conductors, is

$$B_y = \mu H_y = \frac{\mu y I}{2\pi r^2} \text{ Wb/m}^2 \tag{2.12}$$

where μ is the permeability of the conductor.

Consider now an infinitesimal tubular element of thickness dy and length 1m. The flux in the tubular element $d\phi = B_y\, dy$ webers links the fractional turn ($I_y/I = y^2/r^2$) resulting in flux linkages of

$$d\lambda = \left(\frac{y^2}{r^2}\right) d\phi = \left(\frac{y^2}{r^2}\right)\frac{\mu y I}{2\pi r^2} dy \text{ Wb-T/m} \tag{2.13}$$

Integrating, we get the total internal flux linkages as

$$\lambda_{\text{int}} = \int_0^r \frac{\mu I}{2\pi r^4} y^3 dy = \frac{\mu I}{8\pi} \text{ Wb-T/m} \tag{2.14}$$

For a relative permeability $\mu_r = 1$ (nonmagnetic conductor), $\mu = 4\pi \times 10^{-7}$ H/m; therefore,

$$\lambda_{\text{int}} = \frac{I}{2} \times 10^{-7} \text{ Wb-T/m} \tag{2.15}$$

and

$$L_{\text{int}} = \frac{1}{2} \times 10^{-7} \text{ H/m} \tag{2.16}$$

2.3.2 Flux Linkages due to Flux between Two Points External to Conductor

Figure 2.2 shows two points P_1 and P_2 at distances D_1 and D_2 from a conductor which carries a current of I amperes. As the conductor is far removed from the return current path, the magnetic field external to the conductor is concentric circles around the conductor and therefore all the flux between P_1 and P_2 lines within the concentric cylindrical surfaces are passing through P_1 and P_2.

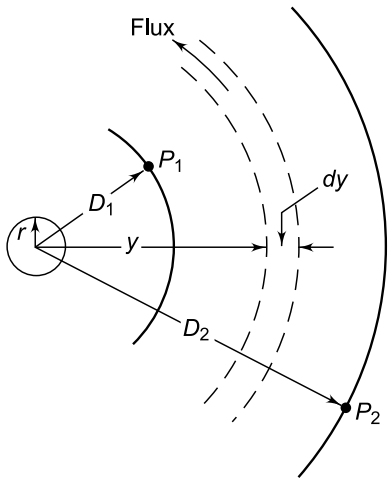


Fig. 2.2 *Flux linkages due to flux between external points P_1, P_2*

Magnetic field intensity at distance y from the conductor is

$$H_y = \frac{I}{2\pi y} \text{ AT/m}$$

The flux dϕ contained in the tubular element of thickness dy is

$$d\phi = \frac{\mu I}{2\pi y} dy \text{ Wb/m length of conductor}$$

The flux dϕ being external to the conductor, links all the current in the conductor which together with the return conductor at infinity forms a single return, such that its flux linkages are given by

$$d\lambda = 1 \times d\phi = \frac{\mu I}{2\pi y} dy$$

Therefore, the total flux linkages of the conductor due to flux between points P_1 and P_2 is

$$\lambda_{12} = \int_{D_1}^{D_2} \frac{\mu I}{2\pi y} \mathrm{d}y = \frac{\mu}{2\pi} I \ln \frac{D_2}{D_1} \text{ Wb-T/m}$$

where ln stands for natural logarithm*.

Since $\mu_r = 1$, $\mu = 4\pi \times 10^{-7}$

$$\therefore \quad \lambda_{12} = 2 \times 10^{-7} I \ln \frac{D_2}{D_1} \text{ Wb-T/m} \tag{2.17}$$

The inductance of the conductor contributed by the flux included between points P_1 and P_2 is then

$$L_{12} = 2 \times 10^{-7} \ln \frac{D_2}{D_1} \text{ H/m} \tag{2.18}$$

or

$$L_{12} = 0.461 \log \frac{D_2}{D_1} \text{ mH/km} \tag{2.19}$$

2.3.3 Flux Linkages due to Flux upto an External Point

Let the external point be at distance D from the centre of the conductor. Flux linkages of the conductor due to external flux (from the surface of the conductor upto the external point) is obtained from Eq. (2.17) by substituting $D_1 = r$ and $D_2 = D$, i.e.,

$$\lambda_{\text{ext}} = 2 \times 10^{-7} I \ln \frac{D}{r} \tag{2.20}$$

Total flux linkages of the conductor due to internal and external flux are

$$\begin{aligned} \lambda &= \lambda_{\text{int}} + \lambda_{\text{ext}} \\ &= \frac{I}{2} \times 10^{-7} + 2 \times 10^{-7} I \ln \frac{D}{r} \\ &= 2 \times 10^{-7} I \left(\frac{1}{4} + \ln \frac{D}{r} \right) \\ &= 2 \times 10^{-7} I \ln \frac{D}{re^{-1/4}} \end{aligned}$$

Let,

$$r' = re^{-1/4} = 0.7788r$$

$$\therefore \quad \lambda = 2 \times 10^{-7} I \ln \frac{D}{r'} \text{ Wb-T/m} \tag{2.21a}$$

Inductance of the conductor due to flux upto an external point is therefore

$$L = 2 \times 10^{-7} \ln \frac{D}{r'} \text{ H/m} \tag{2.21b}$$

Here r' can be regarded as the radius of a fictitious conductor with no internal inductance but the same total inductance as the actual conductor.

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* Throughout the book ln denotes natural logarithm (base e), while log denotes logarithm to base 10.

2.4 INDUCTANCE OF A SINGLE-PHASE TWO-WIRE LINE

Consider a simple two-wire line composed of solid round conductors carrying currents I_1 and I_2 as shown in Fig. 2.3. In a single-phase line,

$$I_1 + I_2 = 0$$

or

$$I_2 = -I_1$$

It is important to note that the effect of earth's presence on magnetic field geometry* is insignificant. This is so because the relative permeability of earth is about the same as that of air and its electrical conductivity is relatively small.

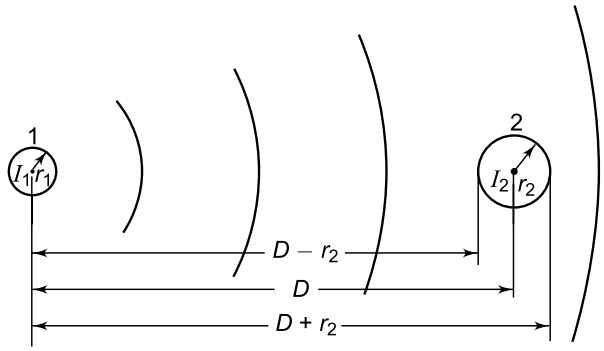


Fig. 2.3 *Single-phase two-wire line and the magnetic field due to current in conductor 1 only*

To start with, let us consider the flux linkages of the circuit caused by current in conductor 1 only. We make three observations in regard to these flux linkages:

1. External flux from r_1 to $(D - r_2)$ links all the current I_1 in conductor 1.
2. External flux from $(D - r_2)$ to $(D + r_2)$ links a current whose magnitude progressively reduces from I_1 to zero along this distance, because of the effect of negative current flowing in conductor 2.
3. Flux beyond $(D + r_2)$ links a net current of zero.

For calculating the total inductance due to current in conductor 1, a simplifying assumption will now be made. If D is much greater than r_1 and r_2 (which is normally the case for overhead lines), it can be assumed that the flux from $(D - r_2)$ to the centre of conductor 2 links all the current I_1 and the flux from the centre of conductor 2 to $(D + r_2)$ links zero current**.

Based on the above assumption, the flux linkages of the circuit caused by current in conductor 1 as per Eq. (2.21a) are

$$\lambda_1 = 2 \times 10^{-7} I_1 \ln \frac{D}{r_1'} \tag{2.22a}$$

The inductance of the conductor due to current in conductor 1 only is then

$$L_1 = 2 \times 10^{-7} \ln \frac{D}{r_1'} \tag{2.22b}$$

Similarly, the inductance of the circuit due to current in conductor 2 is

$$L_2 = 2 \times 10^{-7} \ln \frac{D}{r_2'} \tag{2.23}$$

Using the superposition theorem, the flux linkages and likewise the inductances of the circuit caused by current in each conductor considered separately may be added to obtain the total circuit inductance. Therefore, for the complete circuit

$$L = L_1 + L_2 = 4 \times 10^{-7} \ln \frac{D}{\sqrt{r_1' r_2'}} \text{ H/m} \tag{2.24}$$

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* The electric field geometry will, however, be very much affected as we shall see later while dealing with capacitance.

** Kimbark [10] has shown that the results based on this assumption are fairly accurate even when D is not much larger than r_1 and r_2.

If $r_1' = r_2' = r'$; then

$$L = 4 \times 10^{-7} \ln D/r' \text{ H/m} \quad (2.25a)$$

$$L = 0.921 \log D/r' \text{ mH/km} \quad (2.25b)$$

Transmission lines are infinitely long compared to D in practical situations and therefore the end effects in the above derivation have been neglected.

2.5 CONDUCTOR TYPES

Self-study

So far we have considered transmission lines consisting of single solid cylindrical conductors for forward and return paths. To provide the necessary flexibility for stringing, conductors used in practice are always stranded except for very small cross-sectional areas. Stranded conductors are composed of strands of wire, electrically in parallel, with alternate layers spiralled in opposite direction to prevent unwinding. The total number of strands (N) in concentrically stranded cables with total annular space filled with strands of uniform diameter (d) is given by

$$N = 3x^2 - 3x + 1 \quad (2.26a)$$

where x is the number of layers wherein the single central strand is counted as the first layer. The overall diameter (D) of a stranded conductor is

$$D = (2x - 1)d \quad (2.26b)$$

Aluminium is now the most commonly employed conductor material. It has the advantages of being cheaper and lighter than copper though with less conductivity and tensile strength. Low density and low conductivity result in larger overall conductor diameter, which offers another incidental advantage in high voltage lines. Increased diameter results in reduced electrical stress at conductor surface for a given voltage so that the line is *corona free*. The low tensile strength of aluminium conductors is made up by providing central strands of high tensile strength steel. Such a conductor is known as aluminium conductor steel reinforced (ACSR) and is most commonly used in overhead transmission lines. Figure 2.4 shows the cross-sectional view of an ACSR conductor with 24 strands of aluminium and 7 strands of steel.

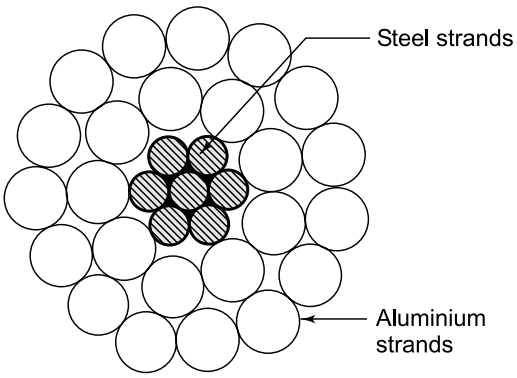


Fig. 2.4 *Cross-sectional view of ACSR-7 steel strands, 24 aluminium strands*

In extra high voltage (EHV) transmission line, *expanded* ACSR conductors are used. These are provided with paper or hessian between various layers of strands so as to increase the overall conductor diameter in an attempt to reduce electrical stress at conductor surface and prevent corona. The most effective way of constructing *corona-free* EHV lines (Ch.19) is to provide several conductors per phase in suitable geometrical configuration. These are known as *bundled conductors* and are a common practice now for EHV lines.

Table 2.1 ACSR Conductor Characteristic (Aluminium Conductor Steel Reinforced)

Sl. No.	*Name*	*Overall Dia (mm)*	*DC Resistance (ohms/km)*	*Current Capacity (Amp) 75 deg. C*
1.	Mole	4.50	2.78	70
2.	Squirrel	6.33	1.394	107

(*Contd*)

Table 2.1 (Contd)

Sl. No.	*Name*	*Overall Dia (mm)*	*DC Resistance (ohms/km)*	*Current Capacity (Amp) 75 deg. C*
3.	Weasel	7.77	0.9291	138
4.	Rabbit	10.05	0.5524	190
5.	Racoon	12.27	0.3712	244
6.	Dog	14.15	0.2792	291
7.	Wolf	18.13	0.1871	405
8.	Lynx	19.53	0.161	445
9.	Panther	21.00	0.139	487
10.	Zebra	28.62	0.06869	737
11.	Dear	29.89	0.06854	756
12.	Moose	31.77	0.05596	836
13.	Bersimis	35.04	0.04242	998

Note: For more technical particulars of ACSR 'Panther', 'Zebra' and 'Moose' conductors, the reader may refer to http://www.upptdcl.org/tech_info/table_14I.html

2.6 FLUX LINKAGES OF ONE CONDUCTOR IN GROUP

As shown in Fig. 2.5, consider a group of n parallel round conductors carrying phasor currents I_1, I_2,..., I_n whose sum equals zero. Distances of these conductors from a remote point P are indicated as D_1, D_2,..., D_n. Let us obtain an expression for the total flux linkages of the ith conductor of the group considering flux upto the point P only.

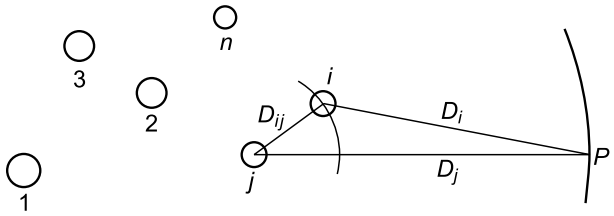


Fig. 2.5 *Arbitrary group of n parallel round conductors carrying currents*

The flux linkages of ith conductor due to its own current I_i (self linkages) are given by [see Eq. (2.21)]

$$\lambda_{ii} = 2 \times 10^{-7} I_i \ln \frac{D_i}{r_i'} \quad \text{Wb-T/m} \tag{2.27}$$

The flux linkages of conductor i due to current in conductor j [refer to Eq. (2.17)] is

$$\lambda_{ij} = 2 \times 10^{-7} I_j \ln \frac{D_j}{D_{ij}} \quad \text{Wb-T/m} \tag{2.28}$$

where D_{ij} is the distance of ith conductor from jth conductor carrying current I_j. From Eq. (2.27) and by repeated use of Eq. (2.28), the total flux linkages of conductor i due to flux upto point P are

$$\lambda_i = \lambda_{i1} + \lambda_{i2} + \cdots + \lambda_{ii} + \cdots + \lambda_{in} = 2 \times 10^{-7} \left(I_1 \ln \frac{D_1}{D_{i1}} + I_2 \ln \frac{D_2}{D_{i2}} + \cdots + I_i \ln \frac{D_i}{r_i'} + \cdots + I_n \ln \frac{D_n}{D_{in}} \right)$$

The above equation can be reorganised as

$$\lambda_i = 2 \times 10^{-7} \left[\left(I_1 \ln \frac{1}{D_{i1}} + I_2 \ln \frac{1}{D_{i2}} + \cdots + I_i \ln \frac{1}{r_i'} + \cdots + I_n \ln \frac{1}{D_{in}} \right) + (I_1 \ln D_1 + I_2 \ln D_2 + \cdots \right.$$

$$\left. + I_i \ln D_i + \cdots + I_n \ln D_n) \right] \tag{2.29}$$

But, $I_n = -(I + I_2 + \cdots + I_{n-1})$.

Substituting for I_n in the second term of Eq. (2.29) and simplifying, we have

$$\lambda_i = 2\times 10^{-7}\left[\left(I_1 \ln\frac{1}{D_{i1}} + I_2 \ln\frac{1}{D_{i2}} + \cdots + I_i \ln\frac{1}{r_i'} + \cdots + I_n \ln\frac{1}{D_{in}}\right)\right.$$

$$\left.+\left(I_1 \ln\frac{D_1}{D_n} + I_2 \ln\frac{D_2}{D_n} + \cdots + I_i \ln\frac{D_i}{D_n}\cdots + I_{n-1} \ln\frac{D_{n-1}}{D_n}\right]\right)$$

In order to account for total flux linkages of conductor i, let the point P now recede to infinity. The terms such as $\ln D_1/D_n$, etc., approach $\ln 1 = 0$. Also for the sake of symmetry, denoting r'_i as D_{ii}, we have

$$\lambda_i = 2\times 10^{-7}\left(I_1 \ln\frac{1}{D_{i1}} + I_2 \ln\frac{1}{D_{i2}} + \cdots I_2 \ln\frac{1}{D_{ii}} + \cdots + I_n \ln\frac{1}{D_{in}}\right)\text{Wb-T/m} \qquad (2.30)$$

2.7 INDUCTANCE OF COMPOSITE CONDUCTOR LINES

We are now ready to study the inductance of transmission lines composed of composite conductors. Figure 2.6 shows such a single-phase line comprising composite conductors A and B with A having n parallel filaments and B having m' parallel filaments. Though the inductance of each filament will be somewhat different (their resistances will be equal if conductor diameters are chosen to be uniform), it is sufficiently accurate to assume that the current is equally divided among the filaments of each composite conductor. Thus, each filament of A is taken to carry a current I/n, while each filament of conductor B carries the return current of $-I/m'$.

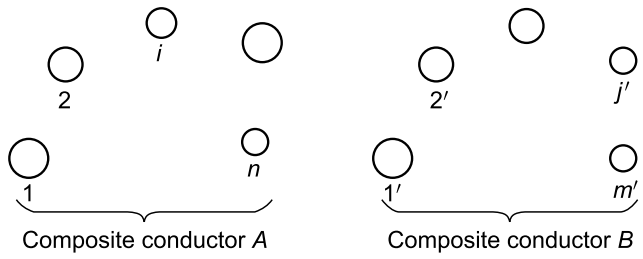


Fig. 2.6 *Single-phase line consisting of two composite conductors*

Applying Eq. (2.30) to filament i of conductor A, we obtain its flux linkages as

$$\lambda_i = 2\times 10^{-7}\frac{I}{n}\left(\ln\frac{1}{D_{i1}} + \ln\frac{1}{D_{i2}} + \cdots + \ln\frac{1}{D_{ii}} + \cdots + \ln\frac{1}{D_{in}}\right) - 2\times 10^{-7}\frac{I}{m'}\left(\ln\frac{1}{D_{i1'}} + \ln\frac{1}{D_{i2'}} + \cdots + \ln\frac{1}{D_{im'}}\right)$$

$$= 2\times 10^{-7} I \ln\frac{(D_{i1'}D_{i2'}\ldots D_{im'})^{1/m'}}{(D_{i1}D_{i2}\ldots D_{ii}\ldots D_{in})^{1/n}}\ \text{Wb-T/m}$$

The inductance of filament i is then

$$L_i = \frac{\lambda_i}{I/n} = 2n\times 10^{-7}\ln\frac{(D_{i1'}\ldots D_{ij'}\ldots D_{im'})^{1/m'}}{(D_{i1}\ldots D_{ii}\ldots D_{in})^{1/n}}\ \text{H/m} \qquad (2.31)$$

The average inductance of the filaments of composite conductor A is

$$L_{\text{avg}} = \frac{L_1 + L_2 + L_3 + \cdots + L_n}{n}$$

Since conductor A is composed of n filaments electrically in parallel, its inductance is

$$L_A = \frac{L_{\text{avg}}}{n} = \frac{L_1 + L_2 + \cdots + L_n}{n^2} \qquad (2.32)$$

Using the expression for filament inductance from Eq. (2.31) in Eq. (2.32), we obtain

$$L_A = 2 \times 10^{-7} \ln \frac{[(D_{1l'} \ldots D_{1j'} \ldots D_{1m'}) \ldots (D_{i1'} \ldots D_{ij'} \ldots D_{im'}) \cdots (D_{nl'} \ldots D_{nj'} \ldots D_{nm'})]^{1/m'n}}{[(D_{1l} \ldots D_{1j} \ldots D_{1n}) \ldots (D_{il} \ldots D_{ii} \ldots D_{in}) \ldots (D_{n1} \ldots D_{ni} \ldots D_{nn})]^{1/n^2}} \text{ H/m} \tag{2.33}$$

The numerator of the argument of the logarithm in Eq. (2.33) is the $m'n$th root of the $m'n$ terms, which are the products of all possible mutual distances from the n filaments of conductor A to m' filaments of conductor B. It is called mutual *geometric mean distance* (mutual GMD) between conductors A and B and is abbreviated as D_m. Similarly, the denominator of the argument of the logarithm in Eq. (2.33) is the n^2th root of n^2 product terms (n sets of n product terms each). Each set of n product term pertains to a filament and consists of $r'(D_{ii})$ for that filament and $(n - 1)$ distances from that filament to every other filament in conductor A. The denominator is defined as the *self geometric mean distance* (self GMD) of conductor A, and is abbreviated as D_{sA}. Sometimes, self GMD is also called *geometric mean radius* (GMR).

In terms of the above symbols, we can write Eq. (2.33) as

$$L_A = 2 \times 10^{-7} \ln \frac{D_m}{D_{sA}} \text{ H/m} \tag{2.34a}$$

$$= 0.461 \log \frac{D_m}{D_{sA}} \text{ mH/km} \tag{2.34b}$$

Note the similarity of the above relation with Eq. (2.22b), which gives the inductance of one conductor of a single-phase line for the special case of two solid, round conductors. In Eq. (2.22b), r_1' is the self GMD of a single conductor and D is the mutual GMD of two single conductors.

The inductance of the composite conductor B is determined in a similar manner, and the total inductance of the line is

$$L = L_A + L_B \tag{2.35}$$

Example 2.1 A conductor is composed of seven identical copper strands, each having a radius r, as shown in Fig. 2.7. Find the self GMD of the conductor.

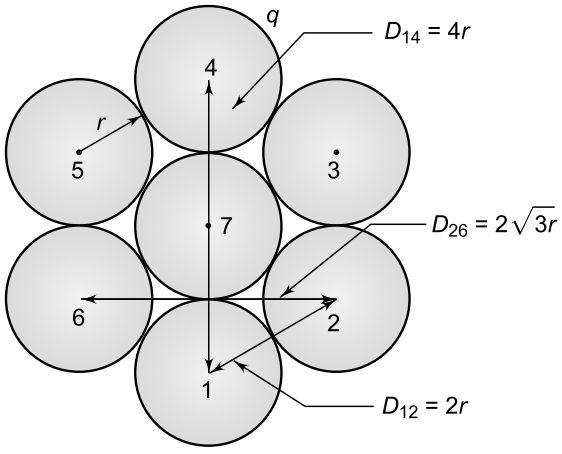


Fig. 2.7 *Cross-section of a seven-strand conductor*

Solution The self GMD of the seven strand conductor is the 49th root of the 49 distances. Thus,

$$D_s = ((r')^7 (D^2_{12} D^2_{26} D_{14} D_{17})^6 (2r)^6)^{1/49}$$

Substituting the values of various distances,

$$D_s = ((0.7788r)^7 (2^2r^2 \times 3 \times 2^2r^2 \times 2^2r \times 2r \times 2r)^6)^{1/49}$$

or

$$D_s = \frac{2r(3(0.7788))^{1/7}}{6^{1/49}} = 2.177r.$$

Example 2.2 The outside diameter of the single layer of aluminium strands of an ACSR conductor shown in Fig. 2.8 is 5.04 cm. The diameter of each strand is 1.68 cm. Determine the 50 Hz reactance at 1 m spacing; neglect the effect of the central strand of steel and advance reasons for the same.

Solution The conductivity of steel being much poorer than that of aluminium and the internal inductance of steel strands being μ-times that of aluminium strands, the current conducted by the central strands of steel can be assumed to be zero.

Diameter of steel strand = $5.04 - 2 \times 1.68 = 1.68$ cm

Thus, all strands are of the same diameter, say d. For the arrangement of strands as given in Fig. 2.8(a),

$$D_{12} = D_{16} = d$$
$$D_{13} = D_{15} = \sqrt{3}d$$
$$D_{14} = 2d$$
$$D_s = \left(\left[\left(\frac{d'}{2}\right)d^2(\sqrt{3}\,d)^2(\sqrt{2}\,d)\right]^6\right)^{1/36}$$

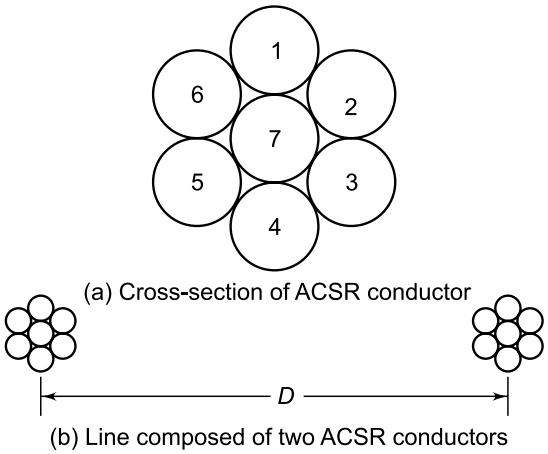


Fig. 2.8

Substituting $d' = 0.7788d$ and simplifying

$$D_s = 1.155d = 1.155 \times 1.68 = 1.93 \text{ cm}$$
$$D_m \simeq D \quad \text{since} \quad D \gg d$$

Now, the inductance of each conductor is

$$L = 0.461 \log \frac{100}{1.93} = 0.789 \text{ mH/km}$$

Loop inductance = $2 \times 0.789 = 1.578$ mH/km

Loop reactance = $1.578 \times 314 \times 10^{-3} = 0.495$ ohms/km.

Example 2.3 The arrangement of conductors of a single-phase transmission line is shown in Fig. 2.9, wherein the forward circuit is composed of three solid wires 2.5 mm in radius and the return circuit of two-wires of radius 5 mm placed symmetrically with respect to the forward circuit. Find the inductance of each side of the line and that of the complete line.

Solution The mutual GMD between sides A and B is

$$D_m = ((D_{14}D_{15})(D_{24}D_{25})(D_{34}D_{35}))^{1/6}$$

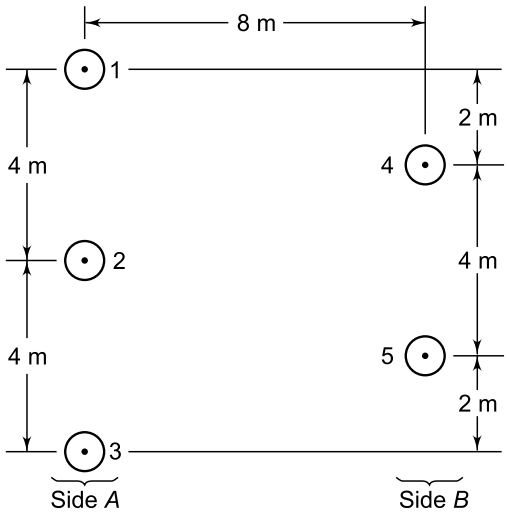


Fig. 2.9 *Arrangement of conductors for Example 2.3*

From the figure it is obvious that

$$D_{14} = D_{24} = D_{25} = D_{35} = \sqrt{68} \text{ m}$$
$$D_{15} = D_{34} = 10 \text{ m}$$
$$D_m = (68^2 \times 100)^{1/6} = 8.8 \text{ m}$$

The self GMD for side A is

$$D_{sA} = ((D_{11}D_{12}D_{13})(D_{21}D_{22}D_{23})(D_{31}D_{32}D_{33}))^{1/9}$$

Here,

$$D_{11} = D_{22} = D_{33} = 2.5 \times 10^{-3} \times 0.7788 \text{ m}$$

Substituting the values of various interdistances and self distances in D_{sA}, we get

$$D_{sA} = ((2.5 \times 10^{-3} \times 0.7788)^3 \times 4^4 \times 8^2)^{1/9}$$
$$= 0.367 \text{ m}$$

Similarly,

$$D_{sB} = ((5 \times 10^{-3} \times 0.7788)^2 \times 4^2)^{1/4}$$
$$= 0.125 \text{ m}$$

Substituting the values of D_m, D_{sA} and D_{sB} in Eq. (2.25b), we get the various inductances as

$$L_A = 0.461 \log \frac{8.8}{0.367} = 0.635 \text{ mH/km}$$

$$L_B = 0.461 \log \frac{8.8}{0.125} = 0.85 \text{ mH/km}$$

$$L = L_A + L_B = 1.485 \text{ mH/km}$$

If the conductors in this problem are each composed of seven identical strands as in Example 2.1, the problem can be solved by writing the conductor self distances as

$$D_{ii} = 2.177 r_i$$

where r_i is the strand radius.

2.8 INDUCTANCE OF THREE-PHASE LINES

So far we have considered only single-phase lines. The basic equations developed can, however, be easily adapted to the calculation of the inductance of three-phase lines. Figure 2.10 shows the conductors of a three-phase line with unsymmetrical spacing.

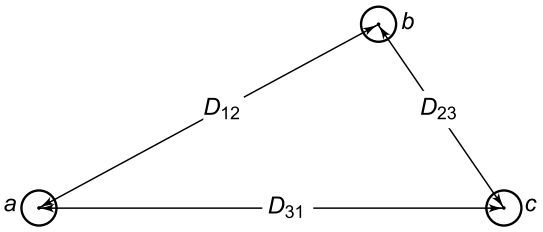


Fig. 2.10 *Cross-sectional view of a three-phase line with unsymmetrical spacing*

Assume that there is no neutral wire, so that

$$I_a + I_b + I_c = 0$$

Unsymmetrical spacing causes the flux linkages and therefore the inductance of each phase to be different resulting in unbalanced receiving-end voltages even when sending-end voltages and line currents are balanced. Also, voltages will be induced in adjacent communication lines even when line currents are balanced. This problem is tackled by exchanging the positions of the conductors at regular intervals along the line such that each conductor occupies the original position of every other conductor over an equal distance. Such an exchange of conductor positions is called *transposition*. A complete transposition cycle

is shown in Fig. 2.11. This arrangement causes each conductor to have the same average inductance over the transposition cycle. Over the length of one transposition cycle, the total flux linkages and net voltage induced in a nearby telephone line is therefore zero.

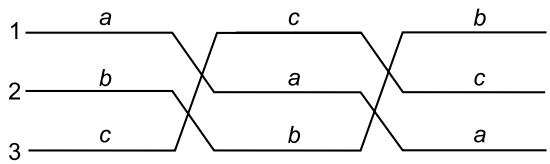


Fig. 2.11 *A complete transposition cycle*

To find the average inductance of each conductor of a transposed line, the flux linkages of the conductor are found for each position it occupies in the transposed cycle. Applying Eq. (2.30) to conductor *a* of Fig. 2.11, for section 1 of the transposition cycle wherein *a* is in position 1, *b* is in position 2 and *c* is in position 3, we get

$$\lambda_{a1} = 2 \times 10^{-7} \left(I_a \ln \frac{1}{r_a'} + I_b \ln \frac{1}{D_{12}} + I_c \ln \frac{1}{D_{31}} \right) \text{ Wb-T/m}$$

For the second section,

$$\lambda_{a2} = 2 \times 10^{-7} \left(I_a \ln \frac{1}{r_a'} + I_b \ln \frac{1}{D_{23}} + I_c \ln \frac{1}{D_{12}} \right) \text{ Wb-T/m}$$

For the third section,

$$\lambda_{a3} = 2 \times 10^{-7} \left(I_a \ln \frac{1}{r_a'} + I_b \ln \frac{1}{D_{13}} + I_c \ln \frac{1}{D_{23}} \right) \text{ Wb-T/m}$$

Average flux linkages of conductor *a* are

$$\lambda_a = \frac{\lambda_{a1} + \lambda_{a2} + \lambda_{a3}}{3} = 2 \times 10^{-7} \left(I_a \ln \frac{1}{r_a'} + I_b \ln \frac{1}{(D_{12}D_{23}D_{31})^{1/3}} + I_c \ln \frac{1}{(D_{12}D_{23}D_{31})^{1/3}} \right)$$

But $I_b + I_c = -I_a$; hence

$$\lambda_a = 2 \times 10^{-7} I_a \ln \frac{(D_{12}D_{23}D_{31})^{1/3}}{r_a'}.$$

Let,

$$D_{eq} = (D_{12}D_{23}D_{31})^{1/3} = \text{equivalent equilateral spacing}$$

Then,

$$L_a = 2 \times 10^{-7} \ln \frac{D_{eq}}{r_a'} = 2 \times 10^{-7} \ln \frac{D_{eq}}{r_a'} \text{ H/m} \tag{2.36}$$

This is the same relation as Eq. (2.34a) where $D_m = D_{eq}$, the mutual GMD between the three-phase conductors. If $r_a = r_b = r_c$, we have

$$L_a = L_b = L_c$$

It is not the present practice to transpose the power lines at regular intervals. However, an interchange in the position of the conductors is made at switching stations to balance the inductance of the phases. For all practical purposes the dissymmetry can be neglected and the inductance of an untransposed line can be taken equal to that of a transposed line.

If the spacing is equilateral, then

$$D_{eq} = D$$

and

$$L_a = 2 \times 10^{-7} \ln \frac{D}{r_a'} \text{ H/m} \tag{2.37}$$

If $r_a = r_b = r_c$, it follows from Eq. (2.37) that

$$L_a = L_b = L_c$$

Example 2.4 Show that over the length of one transposition cycle of a power line, the total flux linkages of a nearby telephone line are zero, for balanced three-phase currents.

Solution Referring to Fig. 2.12, the flux linkages of the conductor t_1 of the telephone line are

$$\lambda_{t1} = 2 \times 10^{-7} \left(I_a \ln \frac{1}{D_{a1}} + \mathrm{l}_b \ln \frac{1}{D_{b1}} + I_c \ln \frac{1}{D_{c1}} \right) \text{Wb-T/m} \tag{2.38}$$

Similarly,

$$\lambda_{t2} = 2 \times 10^{-7} \left(I_a \ln \frac{1}{D_{a2}} + I_b \ln \frac{1}{D_{b2}} + I_c \ln \frac{1}{D_{c2}} \right) \text{Wb-T/m} \tag{2.39}$$

The net flux linkages of the telephone line are

$$\lambda_t = \lambda_{t1} - \lambda_{t2} = 2 \times 10^{-7} \left(I_a \ln \frac{D_{a2}}{D_{a1}} + I_b \ln \frac{D_{b2}}{D_{b1}} + I_c \ln \frac{D_{c2}}{D_{c1}} \right) \text{ Wb-T/m} \tag{2.40}$$

The emf induced in the telephone line loop is

$$E_t = 2\pi f \lambda_t \text{ V/m}$$

Under balanced load conditions, λ_t is not very large because there is a cancellation to a great extent of the flux linkages due to I_a, I_b and I_c. Such cancellation does not take place with harmonic currents which are multiples of three and are therefore in phase. Consequently, these frequencies, if present, may be very troublesome.

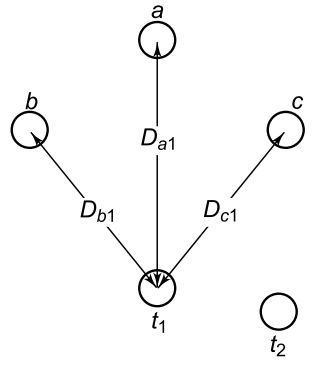


Fig. 2.12 *Effect of transposition on induced voltage of a telephone line*

If the power line is fully transposed with respect to the telephone line

$$\lambda_{t1} = \frac{\lambda_{t1}(\mathrm{I}) + \lambda_{t1}(\mathrm{II}) + \lambda_{t1}(\mathrm{III})}{3}$$

where $\lambda_{t1}(\mathrm{I})$, $\lambda_{t1}(\mathrm{II})$ and $\lambda_{t1}(\mathrm{III})$ are the flux linkages of the telephone line t_1 in the three transposition sections of the power line.

Writing for $\lambda_{t1}(\mathrm{I})$, $\lambda_{t2}(\mathrm{II})$ and $\lambda_{t3}(\mathrm{III})$ by repeated use of Eq. (2.38), we have

$$\lambda_{t1} = 2 \times 10^{-7} (I_a + I_b + I_c) \ln \frac{1}{(D_{a1} D_{b1} D_{c1})^{1/3}}$$

Similarly,

$$\lambda_{t2} = \frac{\lambda_{t2}(\mathrm{I}) + \lambda_{t2}(\mathrm{II}) + \lambda_{t2}(\mathrm{III})}{3}$$

$$= 2 \times 10^{-7} (I_a + I_b + I_c) \ln \frac{1}{(D_{a2} D_{b2} D_{c2})^{1/3}}$$

$$\therefore \quad \lambda_t = 2 \times 10^{-7} (I_a + I_b + I_c) \ln \frac{(D_{a2} D_{b2} D_{c2})^{1/3}}{(D_{a1} D_{b2} D_{c3})^{1/3}} \tag{2.41}$$

If $I_a + I_b + I_c = 0$, $\lambda_t = 0$, i.e., voltage induced in the telephone loop is zero over one transposition cycle of the power line.

It may be noted here that the condition $I_a + I_b + I_c = 0$ is not satisfied for—

1. power frequency L-G (line-to-ground fault) currents, where
$$I_a + I_b + I_c = 3I_0$$
2. third and multiple of third harmonic currents under healthy condition, where,
$$I_a(3) + I_b(3) + I_c(3) = 3I(3)$$
$$\therefore \quad E_t(3) = 6\pi f \lambda_t(3)$$

The harmonic line currents are troublesome in two ways:

1. Induced emf is proportional to the frequency.
2. Higher frequencies come within the audible range.

Thus, there is a need to avoid the presence of such harmonic currents on power line from considerations of the performance of nearby telephone lines.

It has been shown above that voltage induced in a telephone line running parallel to a power line is reduced to zero, if the power line is transposed and provided, it carries balanced currents. It was also shown that power line transposition is ineffective in reducing the induced telephone line voltage when power line currents are unbalanced or when they contain third harmonics. Power line transposition apart from being ineffective introduces mechanical and insulation problems. It is, therefore, easier to eliminate induced voltages by transposing the telephone line instead. In fact, the reader can easily verify that even when the power line currents are unbalanced or when they contain harmonics, the voltage induced over complete transposition cycle (called a *barrel*) of a telephone line is zero. Some induced voltage will always be present on a telephone line running parallel to a power line because in actual practice transposition is never completely symmetrical. Therefore, when the lines run parallel over a considerable length, it is a good practice to transpose both power and telephone lines. The two transposition cycles are staggered and the telephone line is transposed over shorter lengths compared to the power line.

Example 2.5 A three-phase, 50 Hz, 15 km long line has four No. 4/0 wires (1 cm dia) spaced horizontally 1.5 m apart in a plane. The wires in order are carrying currents I_a, I_b and I_c, and the fourth wire, which is a neutral, carries zero current. The currents are:

$$I_a = -30 + j50 \text{ A}$$
$$I_b = -25 + j55 \text{ A}$$
$$I_c = 55 - j105 \text{ A}$$

The line is untransposed.

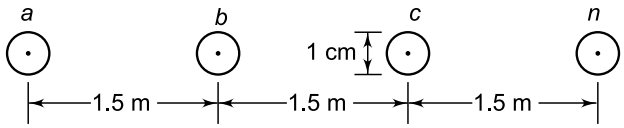


Fig. 2.13 *Arrangement of conductors for Example 2.5*

(a) From the fundamental consideration, find the flux linkages of the neutral. Also, find the voltage induced in the neutral wire.

(b) Find the voltage drop in each of the three-phase wires.

Solution

(a) From Fig. 2.13,

$$D_{an} = 4.5 \text{ m}, D_{bn} = 3 \text{ m}, D_{cn} = 1.5 \text{ m}$$

Flux linkages of the neutral wire n are

$$\lambda_n = 2 \times 10^{-7}\left(I_a \ln \frac{1}{D_{an}} + I_b \ln \frac{1}{D_{bn}} + I_c \ln \frac{1}{D_{cn}} \right) \text{ Wb-T/m}$$

Substituting the values of D_{an}, D_{bn} and D_{cn}, and simplifying, we get

$$\lambda_n = -2 \times 10^{-7} (1.51\, I_a + 1.1\, I_b + 0.405\, I_c) \text{ Wb-T/m}$$

Since $I_c = -(I_a + I_b)$ (this is easily checked from the given values),

$$\lambda_n = -2 \times 10^{-7}(1.105 I_a + 0.695 I_b) \text{ Wb-T/m}$$

The voltage induced in the neutral wire is then

$$V_n = j\omega\lambda_n \times 15 \times 10^3 \text{ V}$$
$$= -j314 \times 15 \times 10^3 \times 2 \times 10^{-7}(1.105 I_a + 0.695 I_b) \text{ V}$$

or, $$V_n = -j0.942\,(1.105\, I_a + 0.695\, I_b) \text{ V}$$

Substituting the values of I_a and I_b, and simplifying,

$$V_n = 0.942 \times 106 = 100 \text{ V.}$$

(b) From Eq. (2.30), the flux linkages of the conductor a are

$$\lambda_a = 2 \times 10^{-7}\left(I_a \ln \frac{1}{r_a'} + I_b \ln \frac{1}{D} + I_c \ln \frac{1}{2D} \right) \text{ Wb-T/m}$$

The voltage drop/metre in phase a can be written as

$$\Delta V_a = 2 \times 10^{-7} j\omega \left(I_a \ln \frac{1}{r_a'} + I_b \ln \frac{1}{D} + I_c \ln \frac{1}{2D} \right) \text{ V/m}$$

Since $I_c = -(I_a + I_b)$, and further since $r_a = r_b = r_c = r$, the expression for ΔV_a can be written in simplified form

$$\Delta V_a = 2 \times 10^{-7} j\omega \left(I_a \ln \frac{2D}{r'} + I_b \ln 2 \right) \text{ V/m}$$

Similarly, voltage drop/metre of phases b and c can be written as

$$\Delta V_b = 2 \times 10^{-7} j\omega I_b \ln \frac{D}{r'}$$

$$\Delta V_c = 2 \times 10^{-7} j\omega \left(I_b \ln 2 + I_c \ln \frac{2D}{r'} \right)$$

Using matrix notation, we can present the result in compact form

$$\begin{bmatrix} \Delta V_a \\ \Delta V_b \\ \Delta V_c \end{bmatrix} = 2 \times 10^{-7} j\omega \begin{bmatrix} \ln 2D/r' & \ln 2 & 0 \\ 0 & \ln D/r' & 0 \\ 0 & \ln 2 & \ln 2D/r' \end{bmatrix} \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix}$$

The voltage drop of phase a is calculated below

$$\Delta V_a = j2 \times 10^{-7} \times 314 \times 15 \times 10^3 \left(\ln \frac{300}{0.39} (-30 + j50) + 0.693(-25 + j55) \right)$$

$$= -(348.6 + j204) \text{ V}$$

Example 2.6 A single-phase 50 Hz power line is supported on a horizontal cross-arm. The spacing between the conductors is 3 m. A telephone line is supported symmetrically below the power line as shown in Fig. 2.14. Find the mutual inductance between the two circuits and the voltage induced per kilometre in the telephone line if the current in the power line is 100 A. Assume the telephone line current to be zero (see Sec. 2.13).

Solution Flux linkages of conductor T_1

$$\lambda_{t1} = 2 \times 10^{-7} \left(I \ln \frac{1}{D_1} - I \ln \frac{1}{D_2} \right) = 2 \times 10^{-7} I \ln \frac{D_2}{D_1}$$

Flux linkages of conductor T_2

$$\lambda_{t2} = 2 \times 10^{-7} I \ln \frac{D_1}{D_2}$$

Total flux linkage of the telephone circuit

$$\lambda_t = \lambda_{t1} - \lambda_{t2} = 4 \times 10^{-7} I \ln \frac{D_2}{D_1}$$

$$M_{pt} = 4 \times 10^{-7} \ln \frac{D_2}{D_1} \text{ H/m}$$

$$= 0.921 \log \frac{D_2}{D_1} \text{ mH/km}$$

$$D_1 = (1.1^2 + 2^2)^{1/2} = (5.21)^{1/2}$$

$$D_2 = (1.9^2 + 2^2)^{1/2} = (7.61)^{1/2}$$

$$M_{pt} = 0.921 \log \left(\frac{761}{521} \right)^{1/2} = 0.0758 \text{ mH/km}$$

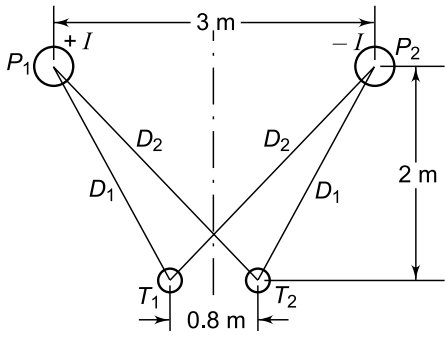


Fig. 2.14 *Power and telephone lines for Example 2.6*

Voltage induced in the telephone circuit $V_t = j\omega M_{pt} I$

$$|V_t| = 314 \times 0.0758 \times 10^{-3} \times 100 = 2.379 \text{ V/km.}$$

2.9 DOUBLE-CIRCUIT THREE-PHASE LINES

It is common practice to build double-circuit three-phase lines so as to increase transmission reliability at somewhat enhanced cost. From the point of view of power transfer from one end of the line to the other (see Sec. 12.3), it is desirable to build the two lines with as low an inductance/phase as possible. In order to achieve this, self GMD (D_s) should be made high and mutual GMD (D_m) should be made low. Therefore, the individual conductors of a phase should be kept as far apart as possible (for high self GMD), while the distance between phases be kept as low as permissible (for low mutual GMD).

Figure 2.15 shows the three sections of the transposition cycle of two parallel circuit three-phase lines with vertical spacing (it is a very commonly used configuration).

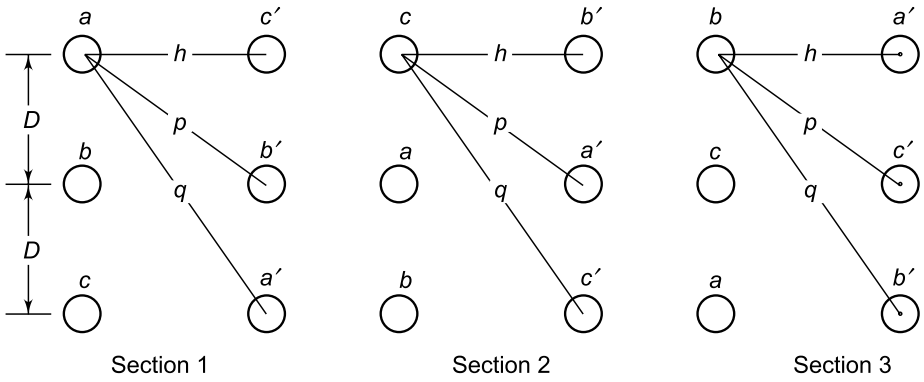


Fig. 2.15

It may be noted here that conductors a and a' in parallel compose phase a and similarly b and b' compose phase b and c and c' compose phase c. In order to achieve high D_s, the conductors of two phases are placed diametrically opposite to each other and those of the third phase are horizontally opposite to each other. (The reader can try other configurations to verify that these will lead to low D_s). Applying the method of GMD, the equivalent equilateral spacing is

$$D_{eq} = (D_{ab}D_{bc}D_{ca})^{1/3} \tag{2.42}$$

where, D_{ab} = mutual GMD between phases a and b in section 1 of the transposition cycle

$= (DpDp)^{1/4} = (Dp)^{1/2}$

D_{bc} = mutual GMD between phases b and c in section 1 of the transposition cycle

$= (Dp)^{1/2}$

D_{ca} = mutual GMD between phases c and a in section 1 of the transposition cycle

$= (2Dh)^{1/2}$

Hence, $D_{eq} = 2^{1/6}D^{1/2}p^{1/3}h^{1/6}$ (2.43)

It may be noted here that D_{eq} remains the same in each section of the transposition cycle, as the conductors of each parallel circuit rotate cyclically, so do D_{ab}, D_{bc} and D_{ca}. The reader is advised to verify this for sections 2 and 3 of the transposition cycle in Fig. 2.15.

Self GMD in section 1 of phase a (i.e., conductors a and a') is

$$D_{sa} = (r'qr'q)^{1/4} = (r'q)^{1/2}$$

Self GMD of phases b and c in section 1 are respectively

$$D_{sb} = (r'hr'h)^{1/4} = (r'h)^{1/2}$$

$$D_{sc} = (r'qr'q)^{1/4} = (r'q)^{1/2}$$

$\therefore$ Equivalent self GMD, $D_s = (D_{sa}D_{sb}D_{sc})^{1/3}$

$$= (r')^{1/2}q^{1/3}h^{1/6} \tag{2.44}$$

Because of the cyclic rotation of conductors of each parallel circuit over the transposition cycle, D_s also remains the same in each transposition section. The reader should verify this for sections 2 and 3 in Fig. 2.15.

The inductance per phase is

$$L = 2 \times 10^{-7} \ln \frac{D_{eq}}{D_s}$$

$$= 2 \times 10^{-7} \ln \frac{2^{1/6}D^{1/2}p^{1/3}h^{1/6}}{(r')^{1/2}q^{1/3}h^{1/6}}$$

$$= 2 \times 10^{-7} \ln \left(2^{1/6} \left(\frac{D}{r'} \right)^{1/2} \left(\frac{p}{q} \right)^{1/3} \right) \text{ H/phase/m} \tag{2.45}$$

The self inductance of each circuit is given by

$$L_s = 2 \times 10^{-7} \ln \frac{(2)^{1/3} D}{r'}$$

Equation (2.45) can now be written as

$$L = \frac{1}{2} \left[2 \times 10^{-7} \ln \frac{(2)^{1/3} D}{r'} + 2 \times 10^{-7} \ln \left(\frac{p}{q} \right)^{2/3} \right] \tag{2.46}$$

$$= \frac{1}{2} (L_s + M)$$

where M is the mutual inductance between the two circuits, i.e.,

$$M = 2 \times 10^{-7} \ln \left(\frac{p}{q} \right)^{2/3}$$

This is a well known result for the two coupled circuits connected in parallel (at similar polarity ends).

If $h \gg D, \left(\frac{p}{q} \right) \to 1$ and $M \to 0$, i.e., the mutual impedance between the circuits becomes zero. Under this condition,

$$L = 1 \times 10^{-7} \ln \frac{3\sqrt{2}\, D}{r'} \tag{2.47}$$

The GMD method, though applied above to a particular configuration of a double circuit, is valid for any configuration as long as the circuits are electrically parallel.

While the GMD method is valid for fully transposed lines, it is commonly applied for untransposed lines and is quite accurate for practical purposes.

2.10 BUNDLED CONDUCTORS

It is economical to transmit large chunks of power over long distances by employing EHV lines. However, the line voltages that can be used are severely limited by the phenomenon of *corona*. Corona, in fact, is the result of ionisation of the atmosphere when a certain field intensity (about 3,000 kV/m at NTP) is reached. Corona discharge causes communication interference and associated power loss which can be severe in bad weather conditions. Critical line voltage for formation of corona can be raised considerably by the use of bundled conductors—a group of two or more conductors per phase. This increase in critical corona voltage is dependent on the number of conductors in the group, the clearance between them and the distance between the groups forming the separate phases*. Reichman [11] has shown that the spacing of conductors in a bundle affects voltage gradient and the optimum spacing is of the order of 8–10 times the conductors diameter, irrespective of the number of conductors in the bundle.

* The bundle usually comprises two, three or four conductors arranged in configurations illustrated in Fig. 2.16. The current will not divide equally among them.

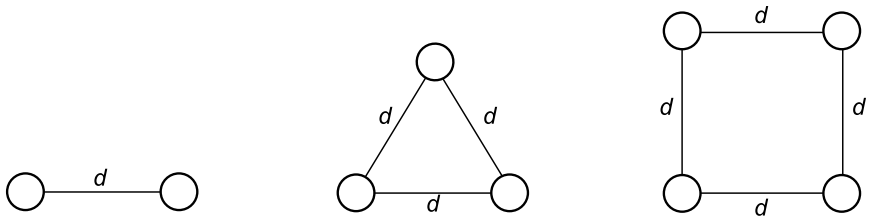


Fig. 2.16 *Configuration of bundled conductors*

Further, because of increased self GMD* line, inductance is reduced considerably with the incidental advantage of increased transmission capacity of the line.

Example 2.7 Find the inductive reactance in ohms per kilometer at 50 Hz of a three-phase bundled conductor line with two conductors per phase as shown in Fig. 2.17. All the conductors are ACSR with radii of 1.725 cm.

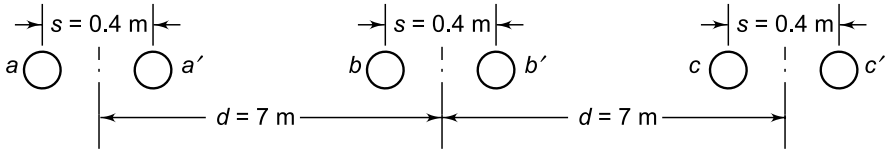


Fig. 2.17 *Bundled conductor three-phase line*

Solution Even though the power lines are not normally transposed (except when they enter and leave a switching station), it is sufficiently accurate to assume complete transposition (of the bundles as well as of the conductors within the bundle) so that the method of GMD can be applied.

The mutual GMD between bundles of phases *a* and *b*

$$D_{ab} = (d\,(d+s)\,(d-s)\,d)^{1/4}$$

Mutual GMD between bundles of phases *b* and *c*

$$D_{bc} = D_{ab} \text{ (by symmetry)}$$

Mutual GMD between bundles of phases *c* and *a*

$$D_{ca} = (2d\,(2d+s\,)\,(2d-s)2d)^{1/4}$$

$$D_{eq} = (D_{ab}D_{bc}D_{ca})^{1/3}$$

$$= (4d^6(d+s)^2(d-s)^2(2d+s)(2d-s))^{1/12}$$

$$= (4(7)^6(7.4)^2(6.6)^2(14.4)(13.6))^{1/12}$$

$$= 8.81 \text{ m}$$

$$D_s = (r'sr's)^{1/4} = (r's)^{1/2} = (0.7788 \times 1.725 \times 10^{-2} \times 0.4)^{1/2}$$

$$= 0.073 \text{ m}$$

Inductive reactance per phase

$$X_L = 314 \times 0.461 \times 10^{-3} \log \frac{8.81}{0.073}$$

$$= 0.301 \text{ ohm/km}$$

* The more the number of conductors in a bundle, the more is the self GMD.

In most cases, it is sufficiently accurate to use the centre to centre distances between bundles rather than mutual GMD between bundles for computing D_{eq}. With this approximation, we have for the example in hand

$$D_{eq} = (7 \times 7 \times 14)^{1/3} = 8.82 \text{ m}$$

$$X_L = 314 \times 0.461 \times 10^{-3} \log \frac{8.82}{0.073}$$

$$= 0.301 \text{ ohm/km}$$

Thus, the approximate method yields almost the same reactance value as the exact method. It is instructive to compare the inductive reactance of a bundled conductor line with an equivalent (on heuristic basis) single conductor line. For the example in hand, the equivalent line will have $d = 7$ m and conductor diameter (for same total cross-sectional area) as $\sqrt{2} \times 1.725$ cm.

$$X_L = 314 \times 0.461 \times 10^{-3} \log \frac{(7 \times 7 \times 14)^{1/3}}{0.7788 \times \sqrt{2} \times 1.725 \times 10^{-3}}$$

$$= 0.531 \text{ ohm/km}$$

This is 76.41% higher than the corresponding value for a bundled conductor line. As already pointed out, lower reactance of a bundled conductor line increases its transmission capacity.

2.11 RESISTANCE

Self-study

Though the contribution of line resistance to series line impedance can be neglected in most cases, it is the main source of line power loss. Thus while considering transmission line economy, the presence of line resistance must be considered.

The effective AC resistance is given by

$$R = \frac{\text{average power loss in conductor in watts}}{I^2} \text{ ohms} \tag{2.48}$$

where I is the rms current in the conductor in amperes. Ohmic or DC resistance is given by the formula

$$R_O = \frac{\rho l}{A} \text{ ohms} \tag{2.49}$$

where, ρ = resistivity of the conductor, ohm-m

l = length, m

A = cross-sectional area, m^2

The effective resistance given by Eq. (2.48) is equal to the DC resistance of the conductor given by Eq. (2.49) only if the current distribution is uniform throughout the conductor.

For small changes in temperature, the resistance increases with temperature in accordance with the relationship

$$R_t = R\,(1 + \alpha_0 t) \tag{2.50}$$

where, R = resistance at temperature 0°C

α_0 = temperature coefficient of the conductor at 0°C

Equation (2.50) can be used to find the resistance R_{t2} at a temperature t_2, if resistance R_{t1} at temperature t_1 is known

$$\frac{R_{t2}}{R_{t1}} = \frac{1/\alpha_0 + t_2}{1/\alpha_0 + t_1} \tag{2.51}$$

Self-study

2.12 SKIN EFFECT AND PROXIMITY EFFECT

The distribution of current throughout the cross-section of a conductor is uniform only when DC is passing through it. On the contrary when AC is flowing through a conductor, the current is non-uniformly distributed over the cross-section in a manner that the current density is higher at the surface of the conductor compared to the current density at its centre. This effect becomes more pronounced as frequency is increased. This phenomenon is called *skin effect*. It causes larger power loss for a given rms AC than the loss when the same value of DC is flowing through the conductor. Consequently, the effective conductor resistance is more for AC than for DC. A qualitative explanation of the phenomenon is given below.

Imagine a solid round conductor (a round shape is considered for convenience only) to be composed of annular filaments of equal cross-sectional area. The flux linking the filaments progressively decreases as we move towards the outer filaments for the simple reason that the flux inside a filament does not link it. The inductive reactance of the imaginary filaments therefore decreases outwards with the result that the outer filaments conduct more AC than the inner filaments (filaments being parallel). With the increase of frequency the non-uniformity of inductive reactance of the filaments becomes more pronounced, so also the non-uniformity of current distribution. For large solid conductors, the skin effect is quite significant even at 50 Hz. The analytical study of skin effect requires the use of Bessel's functions and is beyond the scope of this book.

Apart from the skin effect, non-uniformity of current distribution is also caused by *proximity effect*. Consider a two-wire line as shown in Fig. 2.18. Each line conductor can be divided into sections of equal cross-sectional area (say three sections). Pairs *aa′*, *bb′* and *cc′* can form three loops in parallel. The flux linking loop *aa′* (and therefore its inductance) is the least and it increases somewhat for loops *bb′* and *cc′*. Thus, the density of AC flowing through the conductors is highest at the inner edges (*aa′*) of the conductors and is the least at the outer edges (*cc′*). This type of non-uniform AC current distribution becomes more pronounced as the distance between conductors is reduced. Like skin effect, the non-uniformity of current distribution caused by proximity effect also increases the effective conductor resistance. For normal spacing of overhead lines, this effect is always of negligible order. However, for underground cables where conductors are located close to each other, proximity effect causes an appreciable increase in effective conductor resistance.



Fig. 2.18

Both skin and proximity effects depend upon conductor size, frequency, distance between conductors and permeability of conductor material.

2.13 MAGNETIC FIELD INDUCTION

Transmission lines are used to transmit the bulk amount of power at higher voltages. These lines establish electromagnetic and electrostatic fields of sufficient magnitude in the neighbouring vicinity. These fields induce currents and voltages in the objects that lie in the vicinity of these fields and have a considerable length parallel to the line, i.e., telephone lines, pipe lines etc. (see Example 2.6).

It is a common practice to run the communication lines in parallel or along the same route as the power lines. The induced currents, voltages in the communication lines cause interference with communication service, e.g., damage to apparatus, hazard to person, etc. In the past, researchers have shown that these fields adversely affect the blood composition, growth, immune system and neural functions.

The presence of harmonics and multiple of third harmonics is dangerous for the communication circuits.

2.14 SUMMARY

This chapter has dealt with two important series parameters of transmission lines namely resistance and inductance. Use of bundled conductors, magnetic field induction have also been discussed.

Problems

2.1 Derive the formula for the internal inductance in H/m of a hollow conductor having inside radius r_1 and outside radius r_2 and also determine the expression for the inductance in H/m of a single-phase line consisting of the hollow conductors described above with conductors spaced a distance D apart.

2.2 Calculate the 50 Hz inductive reactance at 1 m spacing in ohms/km of a cable consisting of 12 equal strands around a nonconducting core. The diameter of each strand is 0.25 cm and the outside diameter of the cable is 1.25 cm.

2.3 A concentric cable consists of two thin-walled tubes of mean radii r and R respectively, derive an expression for the inductance of the cable per unit length.

2.4 A single-phase 50 Hz circuit comprises two single-core lead-sheathed cables laid side by side; if the centres of the cables are 0.5 m apart and each sheath has a mean diameter of 7.5 cm, estimate the longitudinal voltage induced per km of sheath when the circuit carries a current of 800 A.

2.5 Two long parallel conductors carry currents of + I and – I. What is the magnetic field intensity at a point P, shown in Fig. P-2.5?

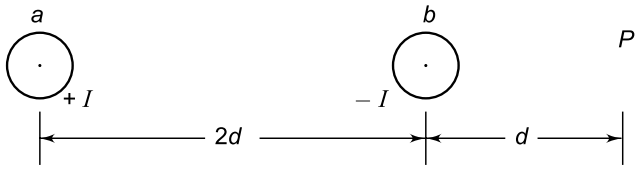


Fig. P-2.5

2.6 Two three-phase lines connected in parallel have self-reactances of X_1 and X_2. If the mutual reactance between them is X_{12}, what is the effective reactance between the two ends of the line?

2.7 A single-phase 50 Hz power line is supported on a horizontal cross-arm. The spacing between conductors is 2.5 m. A telephone line is also supported on a horizontal cross-arm in the same horizontal plane as the power line. The conductors of the telephone line are of solid copper spaced 0.6 m between centres. The distance between the nearest conductors of the two lines is 20 m. Find the mutual inductance between the circuits and the voltage per kilometre induced in the telephone line for 150 A current flowing over the power line.

2.8 A telephone line runs parallel to an untransposed three-phase transmission line, as shown in Fig. P-2.8. The power line carries balanced current of 400 A per phase. Find the mutual inductance between the circuits and calculate the 50 Hz voltage induced in the telephone line per km.

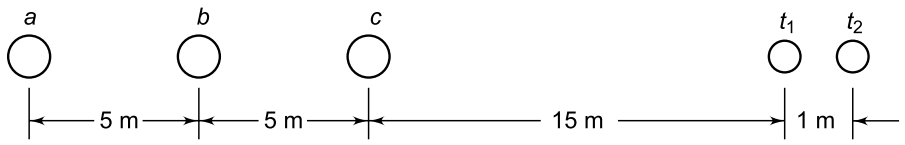


Fig. P-2.8 *Telephone line parallel to a power line*

2.9 A 500 kV line has a bundling arrangement of two conductors per phase as shown in Fig. P-2.9.

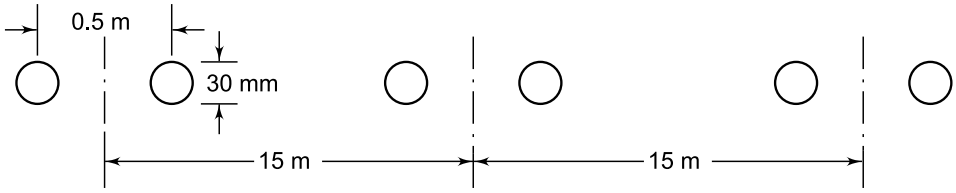


Fig. P-2.9 *500 kV, three-phase bundled conductor line*

Compute the reactance per phase of this line at 50 Hz. Each conductor carries 50% of the phase current. Assume full transposition.

2.10 An overhead line 50 kms in length is to be constructed of conductors 2.56 cm in diameter, for single-phase transmission. The line reactance must not exceed 31.4 ohms. Find the maximum permissible spacing.

2.11 In Fig. P-2.11 which depicts two three-phase circuits on a steel tower, there is symmetry in both the horizontal and vertical centre lines. Let each three-phase circuit be transposed by replacing *a* by *b* and then by *c*, so that the reactances of the three phases are equal and the GMD method of reactance calculations can be used. Each circuit remains on its own side of the tower. Let the self GMD of a single conductor be 1 cm. Conductors *a* and *a′* and other corresponding phase conductors are connected in parallel. Find the reactance per phase of the system.

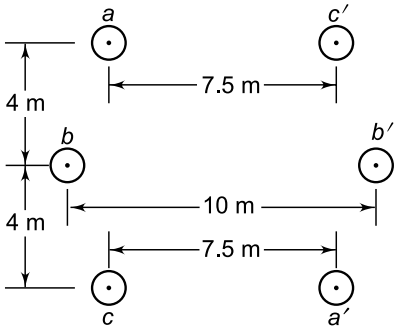


Fig. P-2.11

2.12 A double-circuit three-phase line is shown in Fig. P-2.12. The conductors *a*, *a′*; *b*, *b′* and *c*, *c′* belong to the same phase respectively. The radius of each conductor is 1.5 cm. Find the inductance of the double-circuit line in mH/km/phase.

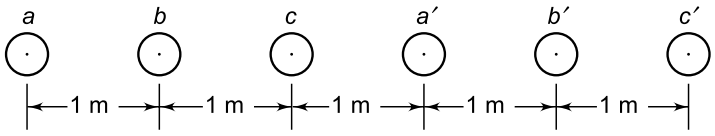


Fig. P-2.12 *Arrangement of conductors for a double-circuit three-phase line*

2.13 A three-phase line with equilateral spacing of 3 m is to be rebuilt with horizontal spacing ($D_{13} = D_{12} = D_{22}$). The conductors are to be fully transposed. Find the spacing between adjacent conductors such that the new line has the same inductance as the original line.

2.14 Find the self GMD of three arrangements of bundled conductors shown in Fig. 2.16 in terms of the total cross-sectional area *A* of conductors (same in each case) and the distance *d* between them.

Multiple Choice Questions

2.1 Bundled conductors in EHV transmission lines results to

(a) Reduce inductance (b) Increase capacitance

(c) Increase inductance (d) Increase resistance

2.2 The internal flux linkage due to internal flux of a conductor is

(a) $I \times 10^{-7}$ Wb-T/m (b) $\frac{I}{4} \times 10^{-7}$ Wb-T/m

(c) $\frac{I}{2} \times 10^{-7}$ Wb-T/m (d) $\frac{1}{2} \times 10^{-7}$ Wb-T/m

2.3 A conductor of radius 'r' will have the fictitious radius 'r' for the inductance calculation

(a) $r' = 2r$ (b) $r' = r$ (c) $r' = 0.7788\, r$ (d) $r' = \frac{r}{0.7788}$

2.4 Aluminium is now most commonly employed conductor material in transmission lines than copper because

(a) It is more conductive (b) Its tensile strength is more
(c) Costlier (d) It is cheaper and lighter

2.5 A conductor is composed of seven identical copper strands each having a radius r, the self GMD of the conductor will be

(a) r (b) $3.177\, r$ (c) $2.177\, r$ (d) none of the above

2.6 The 'skin effect' shows that

(a) The distribution of AC current is uniform through the cross-section of the conductor
(b) Current density is more at the centre of the conductor
(c) Current density is lower at the surface of the conductor
(d) Current density is highest at the surface of the conductor

2.7 Apart from the 'skin effect' the non-uniformity of the current distribution is also caused by

(a) Faraday's effect (b) Bundled conductor (c) Proximity effect (d) Ferranti effect

2.8 The total number of strands (N) in concentrically stranded cable with total annular space filled with strands of uniform diameter is given by (if x is the number of layers wherein the single central strand is counted as the first layer).

(a) $N = 3x^2 + 3x + 1$ (b) $N = 3x^2 - 2x + 1$ (c) $N = 3x^2 - 6x + 1$ (d) $N = 3x^2 - 3x + 1$

2.9 Which of the following is correct:

1. Bundle conductors reduce the corona loss.
2. Bundle conductor increases the inductance and reduces the capacitance.
3. Corona loss causes radio interference in adjoining telephone lines

(a) 1 (b) 1 and 2 (c) 2 (d) 1 and 3

2.10 Bundle conductors are used in transmission lines, the effective capacitance and inductance will respectively

(a) Decrease and increase (b) Increase and decrease
(c) Remain same and increase (d) Decrease and remain same

2.11 Transmission lines are transposed to

(a) Reduce Ferranti effect
(b) Reduce skin effect
(c) Reduce transmission loss
(d) Reduce interference with neighbouring communication lines

2.12 In a transmission line, the distributed constants are

(a) Resistance, inductance (b) Resistance, capacitance
(c) Resistance, inductance, capacitance (d) None of the above

2.13 Skin effect is more pronounced at

(a) High frequency (b) Low frequency (c) Cut-off frequency (d) Half power frequency

2.14 As frequency increases the flux inside the conductor will

(a) Increase (b) Decrease (c) be constant (d) Vanish

2.15 Self GMD is
(a) More than GMR (b) Less than GMR (c) Both are same (d) None of the above

2.16 GMR of a solid conductor is
(a) 0.707 r (b) 0.909 r (c) 0.505 r (d) 0.717 r

2.17 The skin effect of conductor will increase when
(a) Diameter decrease (b) Resistivity decrease
(c) Frequency decrease (d) All of the above

2.18 The fact that a current density is higher at the surface when compared to centre is known as
(a) Skin effect (b) Corona (c) Proximity effect (d) All of the above

2.19 The inductance of single phase two wire line is given by
(a) $4 \times 10^{-7} \ln D/r'$ H/m (b) $0.4 \times 10^{-7} \ln D/r'$ H/m
(c) $4 \times 10^{-7} \ln r'/D$ H/m (d) $0.4 \times 10^{-7} \ln r'/D$ H/m

2.20 ACSR means
(a) Aluminium core steel reinforced (b) Aluminium conductor steel reinforced
(c) Aluminium copper steel reinforced (d) All conductor steel reinfroced

2.21 If the diameter of the conductor is increased
(a) The inductance is increased (b) The inductance is decreased
(c) The resistance is decreased (d) No change in inductance and resistance

2.22 If the self GMD of the conductor is 2.77r, then what is the GMR of the same conductor
(a) 2.77 r (b) 1.385 r (c) 5.54 r (d) (2.77 + 1) r

2.23 The specification of ACSR conductor which having 7 strands of steel and 24 strands of aluminium is
(a) 24/7 (b) 7/24 (c) 7/13 (d) 31/7

2.24 The non-uniformity of current distribution is caused by
(a) Skin effect only (b) Proximity effect (c) Both (a) and (b) (d) Neither (a) nor (b)
(e) Neither (a) or (b)

2.25 Due to proximity effect, the increase in conductor resistance is not negligible in
(a) Underground cable (b) Overhead transmission line
(c) Communication line (d) All of the above

2.26 For a single phase two wire line the diameter 1.213 cm and spacing them is 1.25 m at 50 Hz. The loop inductance is
(a) 22.3×10^{-7} H/m (b) 11.15×10^{-7} H/m (c) 22.3×10^{-4} H/m (d) 11.15×10^{-4} H/m

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Capacitance of Transmission Lines

3

3.1 INTRODUCTION

The capacitance together with conductance forms the shunt admittance of a transmission line. As mentioned earlier, the conductance is the result of leakage over the surface of insulators and is of negligible order. When an alternating voltage is applied to the line, the line capacitance draws a leading sinusoidal current called the *charging current* which is drawn even when the line is open circuited at the far end. The line capacitance being proportional to its length, the charging current is negligible for lines less than 100 km long. For longer lines, the capacitance becomes increasingly important and has to be accounted for.

3.2 ELECTRIC FIELD OF A LONG STRAIGHT CONDUCTOR

Imagine an infinitely long straight conductor far removed from other conductors (including earth) carrying a uniform charge of q coulomb/metre length. By symmetry, the equipotential surfaces will be concentric cylinders, while the lines of electrostatic stress will be radial. The electric field intensity at a distance y from the axis of the conductor is

$$\varepsilon = \frac{q}{2\pi k y} \text{ V/m}$$

where k is the permittivity* of the medium.

As shown in Fig. 3.1, consider two points P_1 and P_2 located at distances D_1 and D_2 respectively from the conductor axis. The potential difference V_{12} (between P_1 and P_2) is given by

$$V_{12} = \oint \varepsilon \, \mathrm{d}\, y = \oint \frac{q}{2\pi k y} \, \mathrm{d}\, y \text{ V}$$

As the potential difference is independent of the path, we choose the path of integration as P_1PP_2 shown in thick line. Since the path PP_2 lies along an equipotential, V_{12} is obtained simply by integrating along P_1P, i.e.,

$$V_{12} = \int_{D_1}^{D_2} \frac{q}{2\pi k y} \mathrm{d}y = \frac{q}{2\pi k} \ln \frac{D_2}{D_1} \quad V \tag{3.1}$$

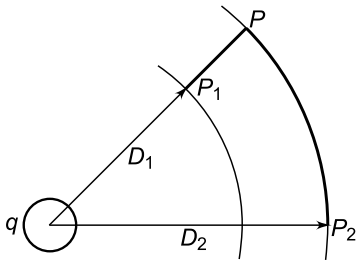


Fig. 3.1 *Electric field of a long straight conductor*

3.3 POTENTIAL DIFFERENCE BETWEEN TWO CONDUCTORS OF A GROUP OF PARALLEL CONDUCTORS

Figure 3.2 shows a group of parallel charged conductors. It is assumed that the conductors are far removed from the ground and are sufficiently removed from each other, i.e., the conductor radii are much smaller than

* In SI units, the permittivity of free space is $k_0 = 8.85 \times 10^{-12}$ F/m. Relative permittivity for air is $k_r = k/k_0 = 1$.

the distances between them. The spacing commonly used in overhead power transmission lines always meets these assumptions. Further, these assumptions imply that the charge on each conductor remains uniformly distributed around its periphery and length. The potential difference between any two conductors of the group can then be obtained by adding the contributions of the individual charged conductors by repeated application of Eq. (3.1). So, the potential difference between conductors a and b (voltage drop from a to b) is

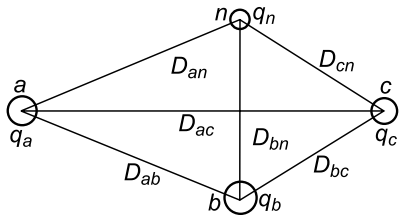


Fig. 3.2 *A group of parallel charged conductors*

$$V_{ab} = \frac{1}{2\pi k}\left(q_a \ln \frac{D_{ab}}{r_a} + q_b \ln \frac{r_b}{D_{ba}} + q_c \ln \frac{D_{cb}}{D_{ca}} + \ldots q_n \ln \frac{D_{nb}}{D_{na}}\right) \text{V} \tag{3.2}$$

Each term in Eq. (3.2) is the potential drop from a to b caused by charge on one of the conductors of the group. Expressions on similar lines could be written for voltage drop between any two conductors of the group.

If the charges vary sinusoidally, so do the voltages (this is the case for AC transmission line), the expression of Eq. (3.2) still applies with charges/metre length and voltages regarded as phasor quantities. Equation (3.2) is thus valid for instantaneous quantities and for sinusoidal quantities as well, wherein all charges and voltages are phasors.

3.4 CAPACITANCE OF A TWO-WIRE LINE

Consider a two-wire line shown in Fig. 3.3 excited from a single-phase source. The line develops equal and opposite sinusoidal charges on the two conductors which can be represented as phasors q_a and q_b so that $q_a = -q_b$.

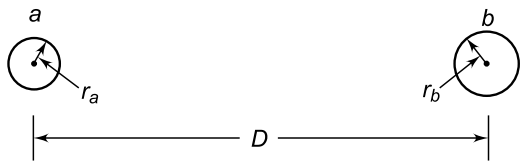


Fig. 3.3 *Cross-sectional view of a two-wire line*

The potential difference V_{ab} can be written in terms of the contributions made by q_a and q_b by use of Eq. (3.2) with associated assumptions (i.e., D/r is large and ground is far away). Thus,

$$V_{ab} = \frac{1}{2\pi k}\left(q_a \ln \frac{D}{r_a} + q_b \ln \frac{r_b}{D}\right) \tag{3.3}$$

Since $q_a = -q_b$, we have

$$V_{ab} = \frac{q_a}{2\pi k} \ln \frac{D^2}{r_a r_b}$$

The line capacitance C_{ab} is then

$$C_{ab} = \frac{q_a}{V_{ab}} = \frac{\pi k}{\ln (D/(r_a r_b)^{1/2})} \text{ F/m length of line} \tag{3.4a}$$

or

$$C_{ab} = \frac{0.0121}{\log (D/(r_a r_b)^{1/2})} \ \mu\text{F/km} \tag{3.4b}$$

If $r_a = r_b = r$,

$$C_{ab} = \frac{0.0121}{\log (D/r)} \ \mu\text{F/km} \tag{3.4c}$$

The associated line charging current is

$$I_c = j\omega C_{ab} V_{ab} \text{ A/km} \tag{3.5}$$

As shown in Figs 3.4(a) and (b) the line-to-line capacitance can be equivalently considered as two equal capacitances in series. The voltage across the lines divides equally between the capacitances such that the neutral point n is at the ground potential. The capacitance of each line to neutral is then given by

$$C_n = C_{an} = C_{bn} = 2C_{ab} = \frac{0.0242}{\log (D/r)} \ \mu\text{F/km} \tag{3.6}$$

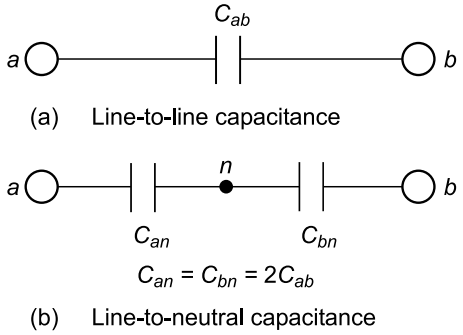


Fig. 3.4

The assumptions inherent in the above derivation are:

1. The charge on the surface of each conductor is assumed to be uniformly distributed, but this is strictly not correct.
   If non-uniformity of charge distribution is taken into account, then

$$C_n = \frac{0.0242}{\log\left(\frac{D}{2r} + \left(\frac{D^2}{4r^2} - 1\right)^{1/2}\right)} \ \mu\text{F/km} \tag{3.7}$$

   If $D/2r \gg 1$, the above expression reduces to that of Eq. (3.6) and the error caused by the assumption of uniform charge distribution is negligible.
2. The cross-section of both the conductors is assumed to be circular, while in actual practice stranded conductors are used. The use of the radius of the circumscribing circle for a stranded conductor causes insignificant error.

3.5 CAPACITANCE OF A THREE-PHASE LINE WITH EQUILATERAL SPACING

Figure 3.5 shows a three-phase line composed of three identical conductors of radius r placed in equilateral configuration.

Using Eq. (3.2) we can write the expressions for V_{ab} and V_{ac} as

$$V_{ab} = \frac{1}{2\pi k}\left(q_a \ln \frac{D}{r} + q_b \ln \frac{r}{D} + q_c \ln \frac{D}{D}\right) \tag{3.8}$$

$$V_{ac} = \frac{1}{2\pi k}\left(q_a \ln \frac{D}{r} + q_b \ln \frac{D}{D} + q_c \ln \frac{r}{D}\right) \tag{3.9}$$

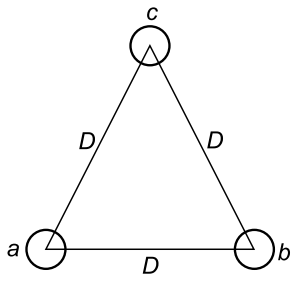


Fig. 3.5 *Cross-section of a three-phase line with equilateral spacing*

Adding Eqs (3.8) and (3.9), we get

$$V_{ab} + V_{ac} = \frac{1}{2\pi k}\left[2q_a \ln \frac{D}{r} + (q_b + q_c) \ln \frac{r}{D}\right] \tag{3.10}$$

Since there are no other charges in the vicinity, the sum of charges on the three conductors is zero. Thus $q_b + q_c = -q_a$, which when substituted in Eq. (3.10) yields

$$V_{ab} + V_{ac} = \frac{3q_a}{2\pi k} \ln \frac{D}{r} \tag{3.11}$$

With balanced three-phase voltages applied to the line, it follows from the phasor diagram of Fig. 3.6 that

$$V_{ab} + V_{ac} = 3V_{an} \tag{3.12}$$

Substituting for $(V_{ab} + V_{ac})$ from Eq. (3.12) in Eq. (3.11), we get

$$V_{an} = \frac{q_a}{2\pi k} \ln \frac{D}{r} \tag{3.13}$$

The capacitance of line to neutral immediately follows as

$$C_n = \frac{q_a}{V_{an}} = \frac{2\pi k}{\ln (D/r)} \tag{3.14a}$$

For air medium ($k_r = 1$),

$$C_n = \frac{0.0242}{\log (D/r)} \ \mu\text{F/km} \tag{3.14b}$$

The line charging current of phase a is

$$I_a \text{ (line charging)} = j\omega C_n V_{an} \tag{3.15}$$

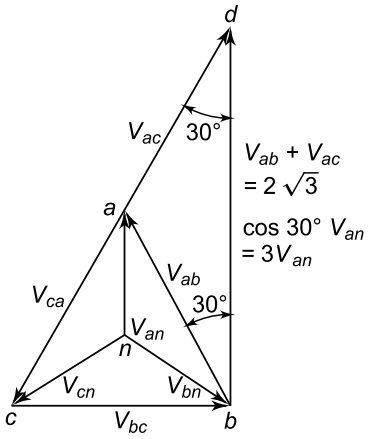


Fig. 3.6 *Phasor diagram of balanced three-phase voltages*

3.6 CAPACITANCE OF A THREE-PHASE LINE WITH UNSYMMETRICAL SPACING

Figure 3.7 shows the three identical conductors of radius r of a three-phase line with unsymmetrical spacing. It is assumed that the line is fully transposed. As the conductors are rotated cyclically in the three sections of the transposition cycle, correspondingly three expressions can be written for V_{ab}. These expressions are:

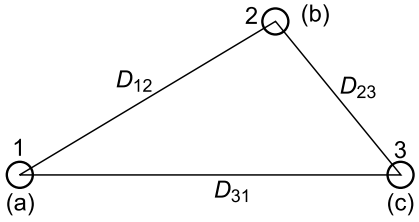


Fig. 3.7 *Cross-section of a three-phase line with asymmetrical spacing (fully transposed)*

For the first section of the transposition cycle

$$V_{ab} = \frac{1}{2\pi k}\left(q_{a1} \ln \frac{D_{12}}{r} + q_{b1} \ln \frac{r}{D_{12}} + q_{c1} \ln \frac{D_{23}}{D_{31}} \right) \tag{3.16a}$$

For the second section of the transposition cycle

$$V_{ab} = \frac{1}{2\pi k}\left(q_{a2} \ln \frac{D_{23}}{r} + q_{b2} \ln \frac{r}{D_{23}} + q_{c2} \ln \frac{D_{31}}{D_{12}} \right) \tag{3.16b}$$

For the third section of the transposition cycle

$$V_{ab} = \frac{1}{2\pi k}\left(q_{a3} \ln \frac{D_{31}}{r} + q_{b3} \ln \frac{r}{D_{31}} + q_{c3} \ln \frac{D_{12}}{D_{23}} \right) \tag{3.16c}$$

If the voltage drop along the line is neglected, V_{ab} is the same in each transposition cycle. On similar lines, three such equations can be written for $V_{bc} = V_{ab} \angle -120°$. Three more equations can be written equating to zero the summation of all line charges in each section of the transposition cycle. From these nine (independent) equations, it is possible to determine the nine unknown charges. The rigorous solution though possible is too involved.

With the usual spacing of conductors, sufficient accuracy is obtained by assuming

$$q_{a1} = q_{a2} = q_{a3} = q_a \,;\, q_{b1} = q_{b2} = q_{b3} = q_b \,;\, q_{c1} = q_{c2} = q_{c3} = q_c \tag{3.17}$$

This assumption of equal charges/unit length of a line in the three sections of the transposition cycle requires, on the other hand, three different values of V_{ab} designated as V_{ab1}, V_{ab2} and V_{ab3} in the three sections. The solution can be considerably simplified by taking V_{ab} as the average of these three voltages, i.e.,

$$V_{ab}\,(\text{avg}) = \frac{1}{3}\,(V_{ab1} + V_{ab2} + V_{ab3})$$

or,
$$V_{ab} = \frac{1}{6\pi k}\left[q_a \ln\left(\frac{D_{12}D_{23}D_{31}}{r^3}\right) + q_b \ln\left(\frac{r^3}{D_{12}D_{23}D_{31}}\right) + q_c \ln\left(\frac{D_{12}D_{23}D_{31}}{D_{12}D_{23}D_{31}}\right)\right]$$

$$= \frac{1}{2\pi k}\left(q_a \ln\frac{D_{eq}}{r} + q_b \ln\frac{r}{D_{eq}}\right) \tag{3.18}$$

where $D_{eq} = (D_{12}D_{23}D_{31})^{1/3}$

Similarly,

$$V_{ac} = \frac{1}{2\pi k}\left(q_a \ln\frac{D_{eq}}{r} + q_c \ln\frac{r}{D_{eq}}\right) \tag{3.19}$$

Adding Eqs (3.18) and (3.19), we get

$$V_{ab} + V_{ac} = \frac{1}{2\pi k}\left(2q_a \ln\frac{D_{eq}}{r} + (q_b + q_c) \ln\frac{r}{D_{eq}}\right) \tag{3.20}$$

As per Eq. (3.12) for balanced three-phase voltages

$$V_{ab} + V_{ac} = 3V_{an}$$

and also $(q_b + q_c) = -q_a$

Use of these relationships in Eq. (3.20) leads to

$$V_{an} = \frac{q_a}{2\pi k} \ln\frac{D_{eq}}{r} \tag{3.21}$$

The capacitance of line to neutral of the transposed line is then given by

$$C_n = \frac{q_a}{V_{an}} = \frac{2\pi k}{\ln\,(D_{eq}/r)} \quad \text{F/m to neutral} \tag{3.22a}$$

For air medium ($k_r = 1$)

$$C_n = \frac{0.0242}{\log\,(D_{eq}/r)} \quad \mu\text{F/km to neutral} \tag{3.22b}$$

It is obvious that for equilateral spacing $D_{eq} = D$, the above (approximate) formula gives the exact result presented earlier.

The line charging current for a three-phase line in phasor form is

$$I_a\,(\text{line charging}) = j\omega C_n V_{an} \quad \text{A/km} \tag{3.23}$$

3.7 EFFECT OF EARTH ON TRANSMISSION LINE CAPACITANCE

So far, in calculating the capacitance of transmission lines, the presence of earth was ignored. The effect of earth on capacitance can be conveniently taken into account by the method of images.

3.7.1 Method of Images Self-study

The electric field of transmission line conductors must conform to the presence of the earth below. The earth for this purpose may be assumed to be a perfectly conducting horizontal sheet of infinite extent which therefore acts like an equipotential surface.

The electric field of two long, parallel conductors charged $+q$ and $-q$ per unit is such that it has a zero potential plane midway between the conductors as shown in Fig. 3.8. If a conducting sheet of infinite dimensions is placed at the zero potential plane, the electric field remains undisturbed. Further, if the conductor carrying charge $-q$ is now removed, the electric field above the conducting sheet stays intact, while that below it vanishes. Using these well known results in reverse, we may equivalently replace the presence of ground below a charged conductor by a fictitious conductor having equal and opposite charge and located as far below the surface of ground as the overhead conductor above it—such a fictitious conductor is the *mirror image* of the overhead conductor. This method of creating the same electric field as in the presence of earth is known as the *method of images* originally suggested by Lord Kelvin.

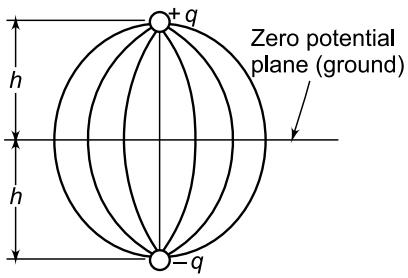


Fig. 3.8 *Electric field of two long, parallel, oppositely charged conductors*

Capacitance of a Single-Phase Line Consider a single-phase line shown in Fig. 3.9. It is required to calculate its capacitance taking the presence of earth into account by the method of images described above. The equation for the voltage drop V_{ab} as determined by the two charged conductors a and b, and their images a' and b' can be written as follows:

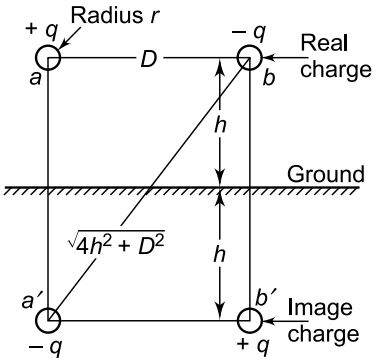


Fig. 3.9 *Single-phase transmission line with images*

$$V_{ab} = \frac{1}{2\pi k}\left[q_a \ln\frac{D}{r} + q_b \ln\frac{r}{D} + q_{a'} \ln\frac{(4h^2 + D^2)^{1/2}}{2h} + q_{b'} \ln\frac{2h}{(4h^2 + D^2)^{1/2}}\right] \tag{3.24}$$

Substituting the values of different charges and simplifying, we get

$$V_{ab} = \frac{q}{\pi k} \ln\frac{2hD}{r(4h^2 + D^2)^{1/2}} \tag{3.25}$$

It immediately follows that

$$C_{ab} = \frac{\pi k}{\ln\dfrac{D}{r(1+(D^2/4h^2))^{1/2}}} \text{ F/m line-to-line} \tag{3.26a}$$

and

$$C_n = \frac{2\pi k}{\ln\dfrac{D}{r(1+(D^2/4h^2))^{1/2}}} \text{ F/m to neutral} \tag{3.26b}$$

It is observed from the above equation that the presence of earth modifies the radius r to $r(1 + (D^2/4h^2))^{1/2}$. For h large compared to D (this is the case normally), the effect of earth on line capacitance is of negligible order.

Capacitance of a Three-Phase Line The method of images can similarly be applied for the calculation of capacitance of a three-phase line, shown in Fig. 3.10. The line is considered to be fully transposed. The conductors *a, b* and *c* carry the charges q_a, q_b and q_c and occupy positions 1, 2 and 3, respectively, in the first section of the transposition cycle. The effect of earth is simulated by image conductors with charges $-q_a$, $-q_b$ and $-q_c$ respectively, as shown.

The equations for the three sections of the transposition cycle can be written for the voltage drop V_{ab} as determined by the three charged conductors and their images. With conductor *a* in position 1, *b* in position 2, and *c* in position 3,

$$V_{ab} = \frac{1}{2\pi k}\left[q_a\left(\ln\frac{D_{12}}{r} - \ln\frac{h_{12}}{h_1}\right) + q_b\left(\ln\frac{r}{D_{12}} - \ln\frac{h_2}{h_{12}}\right) + q_c\left(\ln\frac{D_{23}}{D_{31}} - \ln\frac{h_{23}}{h_{31}}\right)\right] \tag{3.27}$$

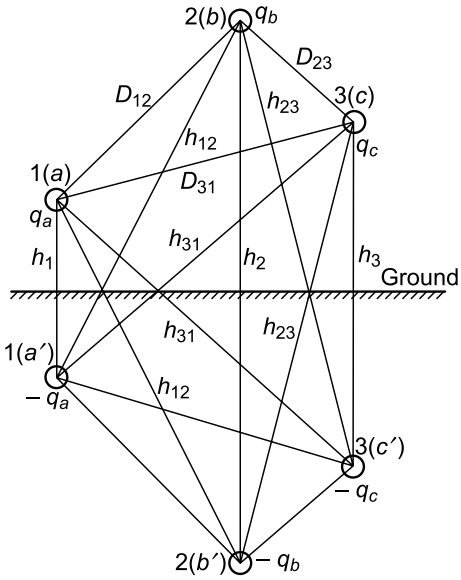


Fig. 3.10 *Three-phase line with images*

Similar equations for V_{ab} can be written for the second and third sections of the transposition cycle. If the fairly accurate assumption of constant charge per unit length of the conductor throughout the transmission cycle is made, the average value of V_{ab} for the three sections of the cycle is given by

$$V_{ab} = \frac{1}{2\pi k}\left[q_a\left(\ln\frac{D_{eq}}{r} - \ln\frac{(h_{12}h_{23}h_{31})^{1/3}}{(h_1h_2h_3)^{1/3}}\right) + q_b\left(\ln\frac{r}{D_{eq}} - \ln\frac{(h_1h_2h_3)^{1/3}}{(h_{12}h_{23}h_{31})^{1/3}}\right)\right] \tag{3.28}$$

where $D_{eq} = (D_{12}D_{23}D_{31})^{1/3}$.

The equation for the average value of the phasor V_{ac} is found in a similar manner. Proceeding on the lines of Sec. 3.6 and using $V_{ab} + V_{ac} = 3V_{an}$ and $q_a + q_b + q_c = 0$, we ultimately obtain the following expression for the capacitance to neutral.

$$C_n = \frac{2\pi k}{\ln\dfrac{D_{eq}}{r} - \ln\left(\dfrac{(h_{12}h_{23}h_{31})^{1/3}}{(h_1h_2h_3)^{1/3}}\right)} \text{ F/m to neutral} \tag{3.29a}$$

or,

$$C_n = \frac{0.0242}{\log\dfrac{D_{eq}}{r} - \log\dfrac{(h_{12}h_{23}h_{31})^{1/3}}{(h_1h_2h_3)^{1/3}}} \ \mu\text{F/km to neutral} \tag{3.29b}$$

Comparing Eqs (3.22a) and (3.29a), it is evident that the effect of earth is to increase the capacitance of a line. If the conductors are high above earth compared to the distances among them, the effect of earth on the capacitance of three-phase lines can be neglected.

Example 3.1 Calculate the capacitance to neutral/km of a single-phase line composed of No. 2 single strand conductors (radius = 0.328 cm) spaced 3 m apart and 7.5 m above the ground. Compare the results obtained by Eqs (3.6), (3.7) and (3.26b).

Solution

1. Neglecting the presence of earth [Eq. (3.6)]

$$C_n = \frac{0.0242}{\log \frac{D}{r}} \quad \mu\text{F/km}$$

$$= \frac{0.0242}{\log \frac{300}{0.328}} = 0.00817 \quad \mu\text{F/km}$$

By the rigorous relationship [(Eq. (3.7)]

$$C_n = \frac{0.0242}{\log\left(\frac{D}{2r} + \left(\frac{D^2}{4r^2} - 1\right)^{1/2}\right)} \quad \mu\text{F/km}$$

Since $\frac{D}{r} = 915$, the effect of non-uniformity of charge distribution is almost negligible.

$\therefore \quad C_n = 0.00817 \quad \mu$F/km.

2. Considering the effect of earth and neglecting non-uniformity of charge distribution [Eq. (3.26b)]

$$C_n = \frac{0.0242}{\log \frac{D}{r(1+(D^2/4h^2))^{1/2}}}$$

$$\frac{D}{r\sqrt{1.04}} = \frac{300}{0.328\sqrt{1.04}} = 897$$

$$C_n = \frac{0.0242}{2.953} = 0.0082 \quad \mu\text{F/km}$$

Note: The presence of earth increases the capacitance by approximately 3 parts in 800.

Example 3.2 A three-phase 50 Hz transmission line has flat horizontal spacing with 3.5 m between adjacent conductors. The conductors are No. 2/0 hard-drawn seven strand copper (outside conductor diameter = 1.05 cm). The voltage of the line is 110 kV. Find the capacitance to neutral and the charging current per kilometre of line.

Solution

$$D_{eq} = (3.5 \times 3.5 \times 7)^{1/3} = 4.4 \text{ m}$$

$$C_n = \frac{0.0242}{\log (D_{eq}/r)} = \frac{0.0242}{\log (440/0.525)}$$

$$= 0.00826 \quad \mu\text{F/km}$$

$$X_n = \frac{1}{\omega C_n} = \frac{10^6}{314 \times 0.00826}$$

$$= 0.384 \times 10^6 \quad \Omega\text{/km to neutral}$$

Charging current $= \frac{V_n}{X_n} = \frac{(110/\sqrt{3}) \times 1000}{0.384 \times 10^6}$

$$= 0.17 \quad \text{A/km}$$

Example 3.3 The six conductors of a double-circuit three-phase line having an overall radius of 0.865 $\times 10^{-2}$ m are arranged as shown in Fig. 3.11. Find the capacitive reactance to neutral and charging current per kilometre per conductor at 110 kV, 50 Hz.

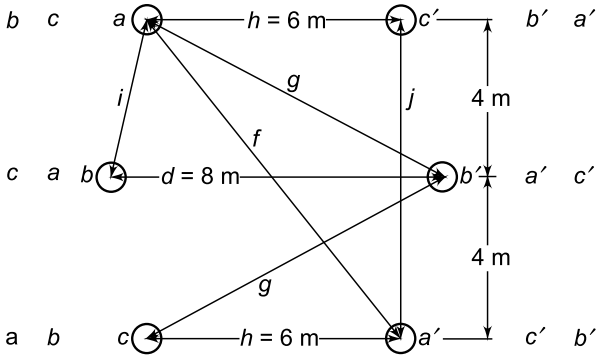


Fig. 3.11 *Cross-section of a double-circuit three-phase line*

Solution As in Sec. 3.6, assume that the charge per conductor on each phase is equal in all the three sections of the transposition cycle. For section *I* of the transposition cycle

$$V_{ab}(\mathrm{I}) = \frac{1}{2\pi k}\left[q_a\left(\ln\frac{i}{r}+\ln\frac{g}{f}\right)+q_b\left(\ln\frac{r}{i}+\ln\frac{d}{g}\right)+q_c\left(\ln\frac{i}{j}+\ln\frac{g}{h}\right)\right] \tag{3.30}$$

For section II of the transposition cycle,

$$V_{ab}(\mathrm{II}) = \frac{1}{2\pi k}\left[q_a\left(\ln\frac{i}{r}+\ln\frac{g}{d}\right)+q_b\left(\ln\frac{r}{i}+\ln\frac{f}{g}\right)+q_c\left(\ln\frac{j}{i}+\ln\frac{h}{g}\right)\right] \tag{3.31}$$

For section III of the transposition cycle,

$$V_{ab}(\mathrm{III}) = \frac{1}{2\pi k}\left[q_a\left(\ln\frac{j}{r}+\ln\frac{h}{f}\right)+q_b\left(\ln\frac{r}{j}+\ln\frac{f}{h}\right)+q_c\left(\ln\frac{i}{j}+\ln\frac{g}{g}\right)\right] \tag{3.32}$$

Average value of V_{ab} over the transposition cycle is given by

$$V_{ab}(\mathrm{avg}) = \frac{1}{6\pi k}\left[q_a\ln\left(\frac{ig\ ig\ jh}{rf\ rd\ rf}\right)+q_b\ln\left(\frac{rd\ rf\ rf}{ig\ ig\ jh}\right)\right]$$

$$= \frac{1}{2\pi k}(q_a-q_b)\ln\left(\frac{i^2g^2jh}{r^3f^2d}\right)^{1/3} \tag{3.33}$$

Similarly,

$$V_{ac}(\mathrm{avg}) = \frac{1}{2\pi k}(q_a-q_c)\ln\left(\frac{i^2g^2jh}{r^3f^2d}\right)^{1/3} \tag{3.34}$$

Now,

$$V_{ab}+V_{ac} = 3V_{an} = \frac{1}{2\pi k}(2q_a-q_b-q_c)\ln\left(\frac{i^2g^2jh}{r^3f^2d}\right)^{1/3} \tag{3.35}$$

$$3V_{an} = \frac{3q_a}{2\pi k}\ln\left(\frac{i^2g^2jh}{r^3f^2d}\right)^{1/3}$$

$$\text{Capacitance to neutral per conductor} = \frac{2\pi k}{\ln\left(\frac{i^2 g^2 jh}{r^3 f^2 d}\right)^{1/3}} \tag{3.36}$$

Total capacitance to neutral for two conductors in parallel

$$C_n = \frac{4\pi k}{\ln\left(\frac{i^2 g^2 jh}{r^3 f^2 d}\right)^{1/3}} \text{ F/m} \tag{3.37}$$

Now, $h = 6$ m; $d = 8$ m; $j = 8$ m. Referring to Fig. 3.12, we can write

$$i = \left[\left(\frac{j}{2}\right)^2 + \left(\frac{d-h}{2}\right)^2\right]^{1/2} = \sqrt{17} \text{ m}$$

$$f = (j^2 + h^2)^{1/2} = 10 \text{ m}$$

$$g = (7^2 + 4^2)^{1/2} = \sqrt{65} \text{ m}$$

Conductor radius (overall) $= 0.865 \times 10^{-2}$ m

Substituting the values for various distances, we have

$$C_n = \frac{4\pi \times 1 \times 8.85 \times 10^{-12} \times 10^6 \times 1000}{\ln\left[\frac{17 \times 65 \times 8 \times 6}{100 \times 8}\left(\frac{100}{0.865}\right)^3\right]^{1/3}} \ \mu\text{F/km}$$

$$= 0.0181 \ \mu\text{F/km}$$

$$\omega C_n = 314 \times 0.0181 \times 10^{-6}$$

$$= 5.68 \times 10^{-6} \ \Omega\text{/km}$$

$$\text{Charging current/phase} = \frac{110 \times 1000}{\sqrt{3}} \times 5.68 \times 10^{-6}$$

$$= 0.361 \text{ A/km}$$

$$\therefore \quad \text{Charging current/conductor} = \frac{0.361}{2} = 0.1805 \text{ A/km}$$

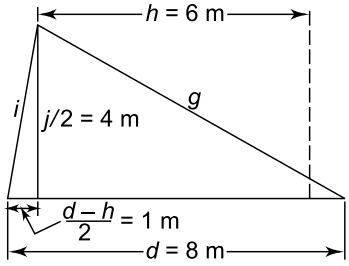
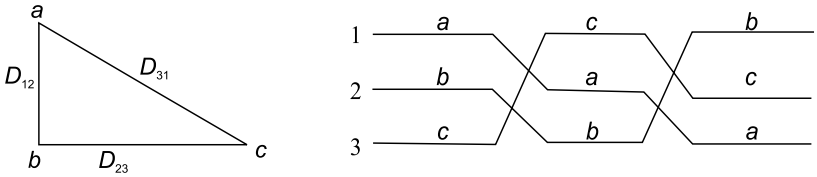


Fig. 3.12

Example 3.4 Derive from the first principles the capacitance per km to neutral of a three-phase overhead transmission line with unsymmetrical spacing of conductors, assuming proper transposition.



Transposition cycle for the Line *a*, *b*, *c*

Fig. 3.13

Solution It is assumed that the charge per unit length of the conductor remains the same in different-positions of the transposition cycle. With this assumption, q_a, q_b, q_c have the same magnitude but 120° appart in phase.

$$q_a = q_a \angle 0° \ ; \ q_b = q_a \angle 120° \ ; \ q_c = q_a \angle 120°$$

Referring to the above figure the potential difference between '*a*' and '*b*' when '*a*' is in position 1, '*b*' in 2 and '*c*' in 3.

$$(V_{ab})_1 = \frac{1}{2\pi\varepsilon}\left[q_a \ln\frac{D_{12}}{r} + q_b \ln\frac{r}{D_{12}} + q_c \ln\frac{D_{23}}{D_{31}}\right] \tag{3.38}$$

When '*a*' in position 2, '*b*' in 3 and '*c*' in 1,

$$(V_{ab})_2 = \frac{1}{2\pi\varepsilon}\left[q_a \ln\frac{D_{23}}{r} + q_b \ln\frac{r}{D_{23}} + q_c \ln\frac{D_{31}}{D_{12}}\right] \tag{3.39}$$

When '*a*' in position 3, '*b*' in 1 and *'c*' in 2,

$$(V_{ab})_3 = \frac{1}{2\pi\varepsilon}\left[q_a \ln\frac{D_{31}}{r} + q_b \ln\frac{r}{D_{31}} + q_c \ln\frac{D_{12}}{D_{23}}\right] \tag{3.40}$$

The average value of potential difference between conductors '*a*' and '*b*' in three positions of the transposition cycle,

$$V_{ab} = \frac{1}{3}\left[(V_{ab})_1 + (V_{ab})_2 + (V_{ab})_3\right]$$

$$V_{ab} = \frac{1}{3}\times\frac{1}{2\pi \in}\left[q_a \ln\frac{D_{12}D_{23}D_{31}}{r^3} + q_b \ln\frac{r^3}{D_{12}D_{23}D_{31}} + q_c \ln\frac{D_{12}D_{23}D_{31}}{D_{12}D_{23}D_{31}}\right]$$

$$= \frac{1}{6\pi\varepsilon}\left[q_a \ln\frac{D_m^3}{r^3} + q_b \ln\frac{r^3}{D_m^3}\right] = \frac{1}{2\pi\varepsilon}\left[q_a \ln\frac{D_m}{r} + q_b \ln\frac{r}{D_m}\right] \tag{3.41}$$

where $D_m = (D_{12}\ D_{23}\ D_{31})^{1/3}$ is the mutual geometric mean distance between the phases.

Similarly, the average potential difference between '*a*' and '*c*'

$$V_{ac} = \frac{1}{2\pi\varepsilon}\left[q_a \ln\frac{D_m}{r} + q_c \ln\frac{r}{D_m}\right] \tag{3.42}$$

But, $V_{ab} + V_{ac} = 3V_{an}$.

Adding Eqs (3.41) and (3.42)

$$3V_{an} = \frac{1}{2\pi\varepsilon}\left[2q_a \ln\frac{D_m}{r} + (q_b + q_c)\ln\frac{r}{D_m}\right]$$

Since, $q_n + q_b + q_c = 0$ in a three phase three wire system,

$$3V_{an} = \frac{1}{2\pi\varepsilon}\left[2q_a \ln\frac{D_m}{r} + (-q_a)\ln\frac{r}{D_m}\right]$$

or,
$$3V_{an} = \frac{1}{2\pi\varepsilon}\left[2q_a \ln\frac{D_m}{r} - q_a \ln\frac{r}{D_m}\right]$$

or,
$$3V_{an} = \frac{1}{2\pi\varepsilon}\left[2q_a \ln\frac{D_m}{r} + q_a \ln\frac{D_m}{r}\right] = \frac{3q_a}{2\pi\varepsilon}\ln\frac{D_m}{r}$$

or,
$$3V_{an} = \frac{q_a}{2\pi\varepsilon}\ln\frac{D_m}{r}.$$

The capacitance of line to neutral is thus,

$$C_n = \frac{q_n}{V_{an}} = \frac{2\pi\varepsilon}{\ln\frac{D_m}{r}} \text{ F/m} = \frac{2\pi\varepsilon}{10^{-3}\ln\frac{D_m}{r}} \text{ F/km.}$$

Example 3.5 A three phase, 50 Hz, 110 kV, transmission line has flat horizontal spacing with 3.5 m between adjacent conductors. The conductor diameter in 1.05 cm.

Find the capacitance to neutral and the charging current per km of line. Assume full transposition.

Solution The equivalent spacing,

$$D_{eq} = (3.5 \times 3.5 \times 7)^{1/3} = 4.5 \text{ m}$$

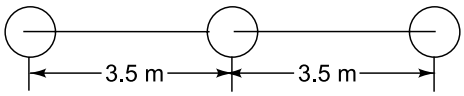


Fig. 3.14

Radius of conductor,

$$r = \frac{1}{2} \times 1.05 \times 10^{-2} = 5.23 \times 10^{-3} \text{ m}$$

Capacitance to neutral, $C_n = \dfrac{2\pi\varepsilon}{\ln\frac{D_m}{r}}$

where, ε = absolute permittivity = $\varepsilon_0\varepsilon_r$

ε_0 = permittivity of space = 8.854×10^{12} F/m

ε_r = relative permittivity = 1 (for aid)

D_m = mutual geometric main distance between the phases

= D_{eq} = 4.5 m

$$C_n = \frac{2\pi\varepsilon_0\varepsilon_r}{\ln\frac{D_m}{r}} = \frac{2\pi \times 8.854 \times 10^{-12}}{\ln\frac{4.5}{5.23 \times 10^{-3}}} = 8.26 \times 10^{-12} \text{ F/m}$$

$$= 8.26 \times 10^9 \text{ F/km} = 8.26 \times 10^3 \ \mu\text{F/km}$$

Capacitive reactance to neutral,

$$X_c = \frac{1}{2\pi f C_n} = \frac{1}{2\pi \times 50 \times 8.26 \times 10^{-9}} = \Omega\text{/km}$$

$$= 3.853 \times 10^5 \ \Omega\text{/km}$$

Charging current, $I_c = \dfrac{V_m}{X_c} = \dfrac{110 \times 10^3/\sqrt{3}}{3.853 \times 10^5} = 0.1648$ A/km

Reactive volt amperes generated by the line = $\dfrac{(V_L)^2}{X_C} = \dfrac{(110 \times 10^3)^2}{3.853 \times 10^5} = 31399$ VAR = 31.399 kVAR.

3.8 METHOD OF GMD (MODIFIED)

A comparison of various expressions for inductance and capacitance of transmission lines [e.g., Eqs (2.22b) and (3.6)] brings out the fact that the two are similar except that in inductance expressions we have to use the fictitious conductor radius $r' = 0.7788r$, while in the expressions for capacitance actual conductor radius r is used. This fact suggests that the method of GMD (Geometric Mean Distance) would be applicable in the calculations for capacitance as well provided it is modified by using the outer conductor radius for finding D_s, the self geometric mean distance.

Example 3.3 can be conveniently solved as under by using the modified GMD method.

For the first section of the transposition cycle mutual GMD is

$$D_{ab} = ((ig)\,(ig))^{1/4} = (ig)^{1/2}$$
$$D_{bc} = (ig)^{1/2}$$
$$D_{ca} = (jh)^{1/2}$$
$$\therefore \quad D_{eq} = (D_{ab}D_{bc}D_{ca})^{1/3} = [(i^2g^2jh)^{1/3}]^{1/2}$$

In the first section of the transposition cycle, self GMD is

$$D_{sa} = (rf\,rf)^{1/4} = (rf)^{1/2}$$
$$D_{sb} = (rd)^{1/2}$$
$$D_{sc} = (rf)^{1/2}$$
$$D_s = (D_{sa}D_{sb}D_{sc})^{1/3} = [(r^3f^2d)^{1/3}]^{1/2}$$

Now,
$$C_n = \frac{2\pi k}{\ln \dfrac{D_{eq}}{D_s}} = \frac{2\pi k}{\ln\left[\left(\dfrac{i^2g^2jh}{r^3f^2d}\right)^{1/3}\right]^{1/2}}$$
$$= \frac{4\pi k}{\ln\left(\dfrac{i^2g^2jh}{r^3f^2d}\right)^{1/3}} \text{ F/m}$$

This result obviously checks with the fundamentally derived expression in Example 3.3.

3.9 BUNDLED CONDUCTORS

A bundled conductor line is shown in Fig. 3.15. The conductors of any one bundle are in parallel, and it is assumed that the charge per bundle divides equally among the conductors of the bundle as $D_{12} >> d$. Also, $D_{12} - d \approx D_{12} + d \approx D_{12}$ for the same reason. The results obtained with these assumptions are fairly accurate for usual spacings. Thus, if the charge on phase a is q_a, the conductors a and a' have a charge of $q_a/2$ each; similarly the charge is equally divided for phases b and c.

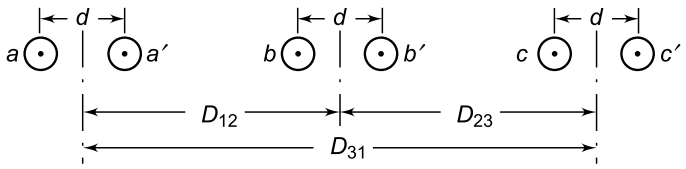


Fig. 3.15 *Cross-section of a bundled conductor three-phase transmission line*

Now, writing an equation for the voltage from conductor a to conductor b, we get

$$V_{ab} = \frac{1}{2\pi k}\left[0.5q_a\left(\ln\frac{D_{12}}{r} + \ln\frac{D_{12}}{d}\right) + 0.5q_b\left(\ln\frac{r}{D_{12}} + \ln\frac{d}{D_{12}}\right) + 0.5q_c\left(\ln\frac{D_{23}}{D_{31}} + \ln\frac{D_{23}}{D_{31}}\right)\right] \tag{3.43}$$

or,

$$V_{ab} = \frac{1}{2\pi k}\left(q_a \ln\frac{D_{12}}{\sqrt{rd}} + q_b \ln\frac{\sqrt{rd}}{D_{12}} + q_c \ln\frac{D_{23}}{D_{31}}\right) \tag{3.44}$$

Considering the line to be transposed and proceeding in the usual manner, the final result will be

$$C_n = \frac{0.0242}{\log\,(D_{eq}/\sqrt{rd})} \ \mu\text{F/km to neutral} \tag{3.45}$$

where $D_{eq} = (D_{12}D_{23}D_{31})^{1/3}$.

It is obvious from Eq. (2.42) that the method of modified GMD is equally valid in this case (as it should be).

3.10 ELECTROSTATIC INDUCTION

Transmission lines as mentioned in Ch. 2, establish both magnetic and electric fields. The electric field also affects the objects that lie in the proximity of the line. Electric fields, related to the voltage of the line, are the main cause of induction to vehicles, buildings and objects of comparable size. The human body is affected adversely by the electric discharges. It has been observed that the current densities in human bodies induced by electric fields are much higher than those induced by magnetic fields.

3.11 SUMMARY

In this chapter, the main shunt parameter of transmission namely capacitance has been thoroughly discussed including the effect of earth and bundled conductors and electrostatic induction is finally dealt with.

Problems

3.1 Derive an expression for the charge (complex) value per metre length of conductor *a* of an untransposed three-phase line shown in Fig. P-3.1. The applied voltage is balanced three-phase, 50 Hz. Take the voltage of phase *a* as reference phasor. All conductors have the same radii. Also find the charging current of phase *a*. Neglect the effect of ground.

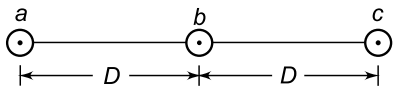


Fig. P-3.1

3.2 A three-phase double-circuit line is shown in Fig. P-3.2. The diameter of each conductor is 2.0 cm. The line is transposed and carries balanced load. Find the capacitance per phase to neutral of the line.

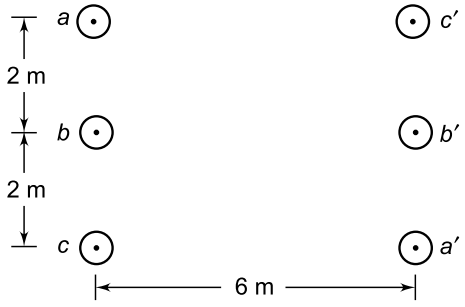


Fig. P-3.2

3.3 A three-phase, 50 Hz overhead line has regularly transposed conductors equilaterally spaced 4 m apart. The capacitance of such a line is 0.01 μF/km. Recalculate the capacitance per kilometre to neutral when the conductors are in the same horizontal plane with successive spacing of 4 m and are regularly transposed.

3.4 Consider the 500 kV, three-phase bundled conductor line as shown in Fig. P-2.9. Find the capacitive reactance to neutral in ohms/km at 50 Hz.

3.5 A three-phase transmission line has flat, horizontal spacing with 2 m between adjacent conductors. The radius of each conductor is 0.25 cm. At a certain instant the charges on the centre conductor and on one of the outside conductors are identical and voltage drop between these identically charged conductors is 775 V. Neglect the effect of ground, and find the value of the identical charge in coulomb/km at the instant specified.

3.6 Find the 50 Hz susceptance to neutral per kilometre of a double-circuit three phase line with transposition as shown in Fig. P-3.6. Given D = 7 m and radius of each of the six conductors is 1.38 cm.

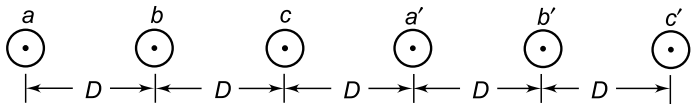


Fig. P-3.6 *Double circuit three-phase line with flat spacing*

3.7 A single conductor power cable has a conductor of No. 2 solid copper (radius = 0.328 cm). Paper insulation separating the conductor from the concentric lead sheath has a thickness of 2.5 mm and a relative permittivity of 3.8. The thickness of the lead sheath is 2 mm. Find the capacitive reactance per kilometre between the inner conductor and the lead sheath.

3.8 Find the capacitance of phase to neutral per kilometre of a three-phase line having conductors of 2 cm diameter placed at the corners of a triangle with sides 5 m, 6 m and 7 m respectively. Assume that the line is fully transposed and carries balanced load.

3.9 Derive an expression for the capacitance per metre length between two long parallel conductors, each of radius, r, with axes separated by a distance D, where $D \gg r$, the insulating medium being air. Calculate the maximum potential difference permissible between the conductors, if the electric field strength between them is not to exceed 25 kV/cm, r being 0.3 cm and D = 35 cm.

Multiple Choice Questions

3.1 The presence of earth in case of overhead lines
(a) Increases capacitance (b) Decreases capacitance
(c) Increases inductance (d) Decreases inductance

3.2 The term self GMD is used to calculate
(a) Capacitance (b) Inductance
(c) Both inductance and capacitance (d) Resistance

3.3 The presence of ground causes the line capacitance to
(a) Increase by about 12% (b) Decrease by about 12%
(c) Increase by about 0.2% (d) None of the above

3.4 If we increase the spacing between the phase conductors, the line capacitance
(a) Decreases (b) Increases (c) Remains the same (d) Not affected

3.5 If we increase the length of transmission line, the charging current
(a) Decreases (b) Increases (c) Remains the same (d) Not affected

3.6 When an alternating voltage is applied to the line, the line capacitance draws a
(a) Leading sinusoidal current (b) Lagging sinusoidal current
(c) A current in phase with voltage (d) None of the above

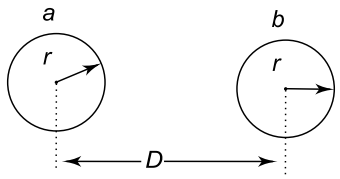
3.7 The capacitance becomes increasingly important for
(a) Short transmission lines (b) Medium transmission lines
(c) Both (a) and (b) (d) Long transmission lines

3.8 An infinitely long straight conductor carries a uniform charge of q coulomb per metre length. If k is permittivity of the medium wherein two points are located at distance D_1 and D_2 respectively from the conductor axis. The potential difference V_{12} (between the two points) is given by

(a) $V_{12} = \dfrac{q}{\pi k} \ln D_2/D_1$ (b) $V_{12} = \dfrac{q}{2\pi k} \ln \dfrac{D_1}{D_2}$

(c) $V_{12} = \dfrac{q}{2\pi k} \ln \dfrac{D_2}{D_1}$ (d) $V_{12} = \dfrac{q}{2\pi k} \log_{10} D_2/D_1$

3.9 Consider a two-wire line shown in figure excited from a single-phase source.



The line capacitance C_{ab} is given by

(a) $C_{ab} = \dfrac{0.0121}{\log(r/D)}$ μF/km (b) $C_{ab} = \dfrac{0.0121}{\log(D/r)}$ μF/km

(c) $C_{ab} = 0.0121 \log(D/r)$ μF/km (d) $C_{ab} = \dfrac{0.0121}{\log(D/r)}$ F/km

3.10 The method of images originally suggested by Lord Kelvin is used in

(a) Calculation of inductance (b) Calculation of resistance

(c) Effect of earth on line capacitance (d) None of the above

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